

Lecture 5: Computational Cameras

Exposure & HDR Imaging

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TAs: Brandon Zhao, Brady Bhalla, Zachary Huang

Course announcements

- HW 1 due this morning @ 10 am
- HW2 released later today
- HW 2 due TUESDAY 4/28 at 10 am on Gradescope!
- After this lecture you should be able to do all questions in the HW except 2 and 3, which should take significantly less time.

Slide credits

Many of these slides/images were borrowed/adapted from:

- Ioannis (Yannis) Gkioulekas (CMU)
- Fredo Durand(MIT)
- Gordon Wetzstein (Stanford)
- Marc Levoy (Stanford/Google)

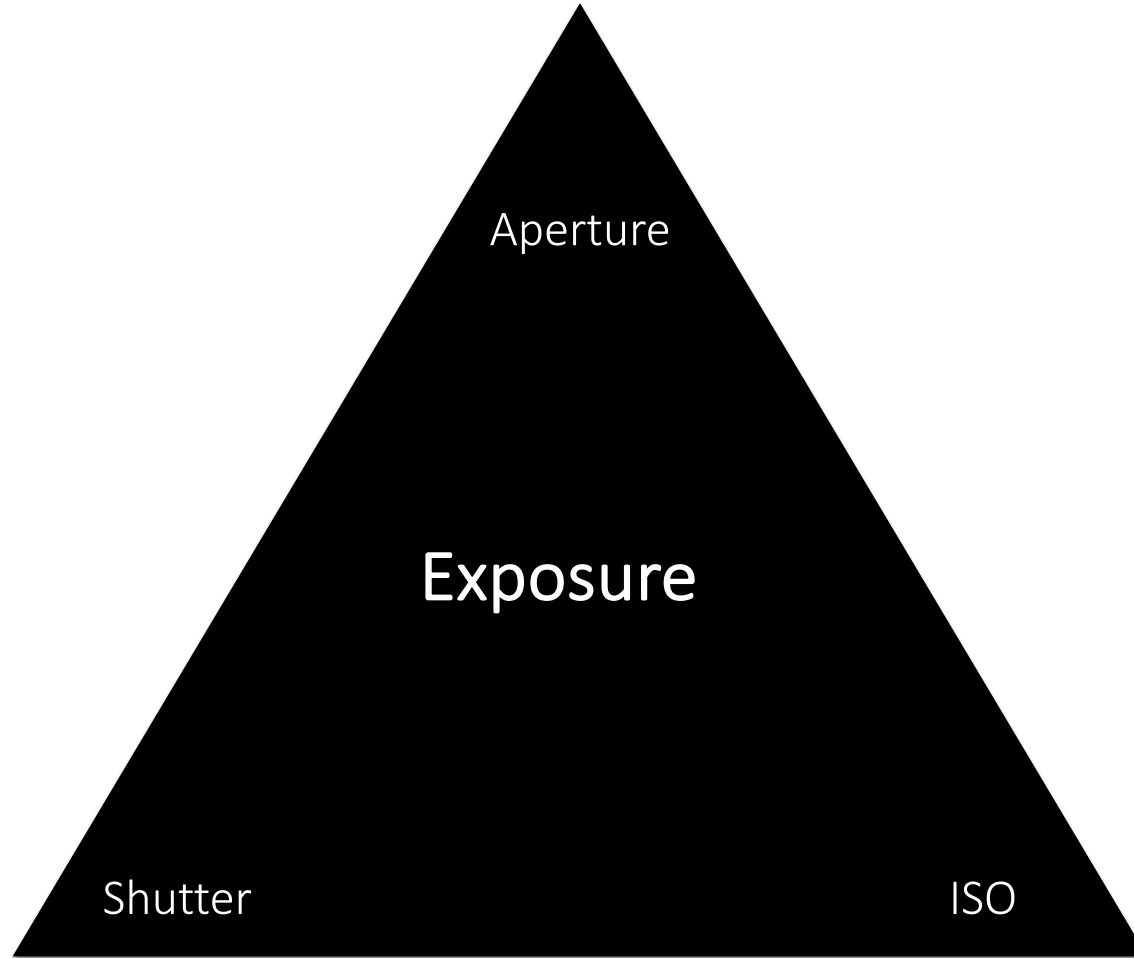
What is exposure?

Roughly speaking, the “brightness” of a captured image given a fixed scene.

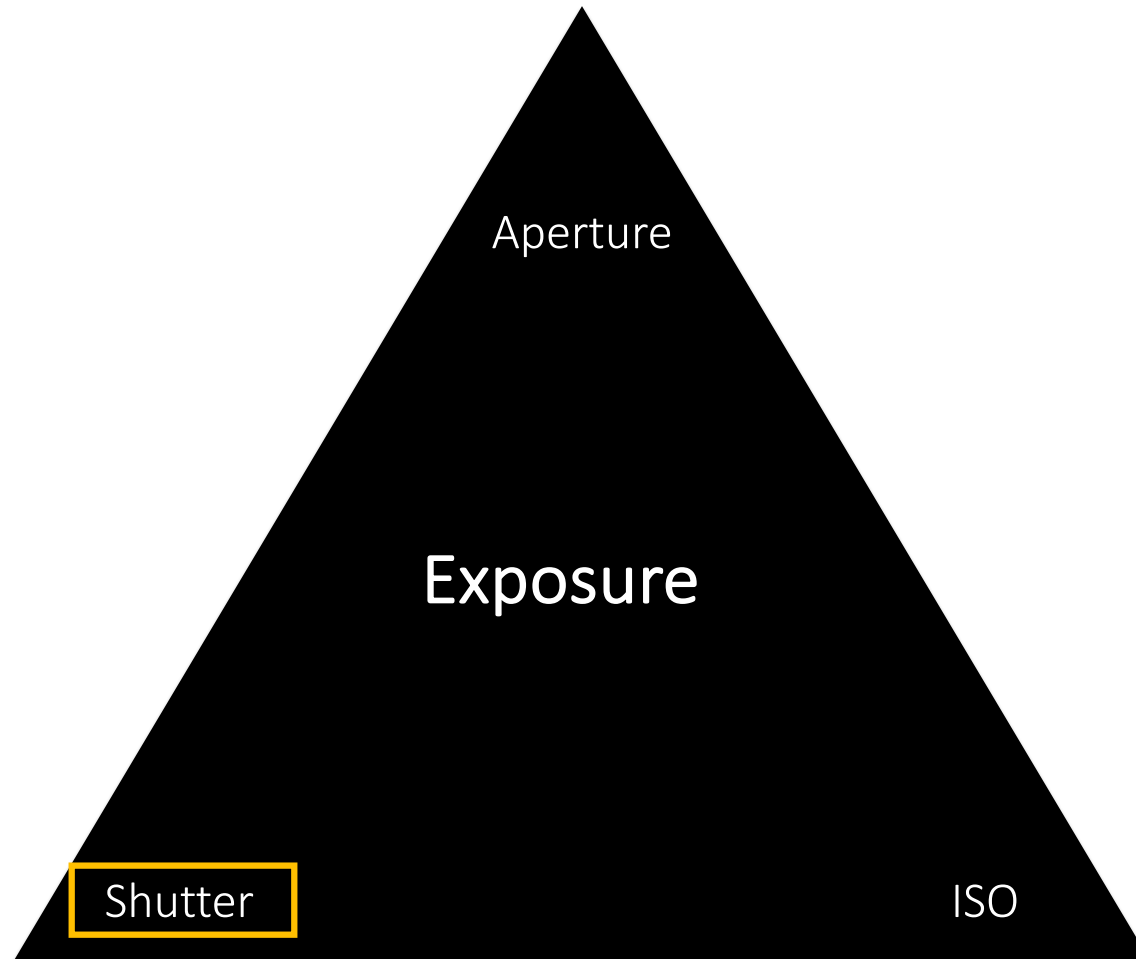
$$\text{Exposure} = \text{Gain} \times \text{Irradiance} \times \text{Time}$$

flux received by a surface per unit area

Exposure controls brightness of image

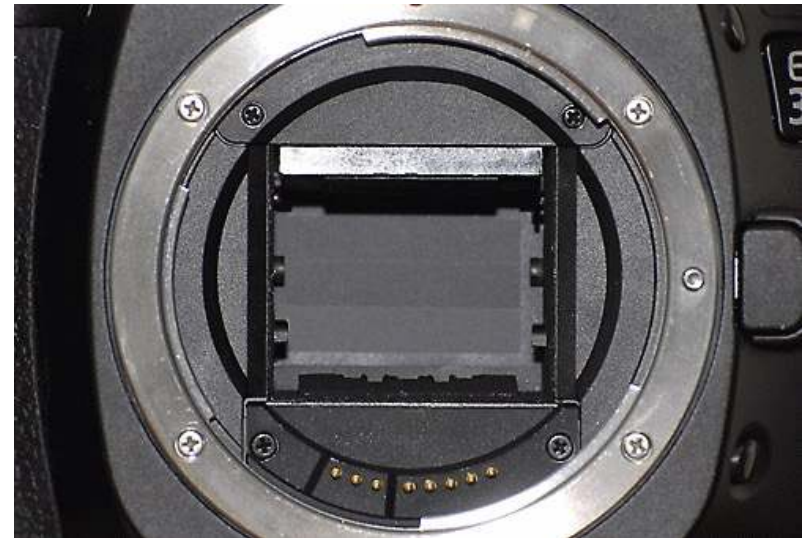
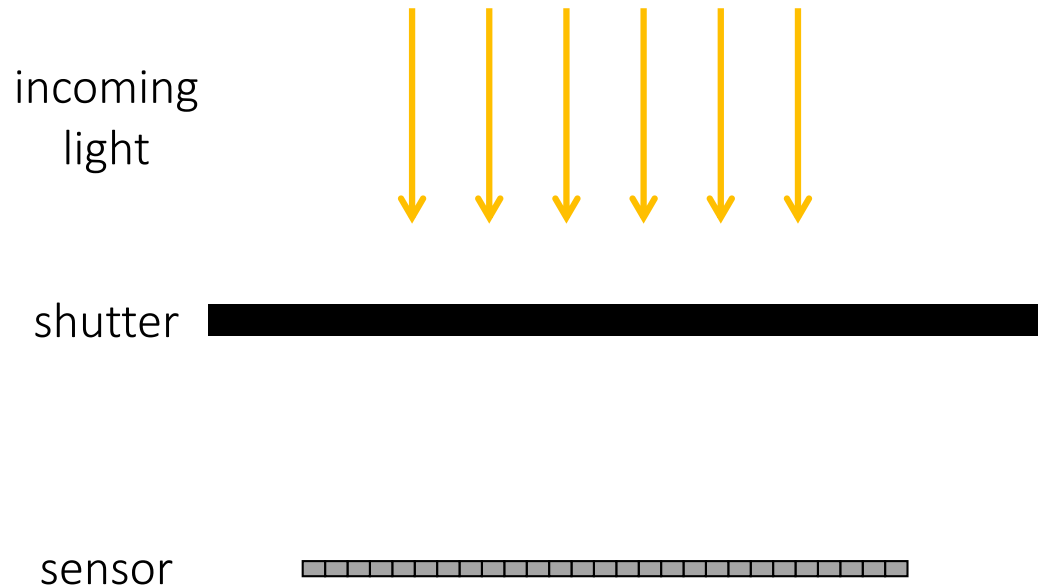


Exposure controls brightness of image



Shutter speed

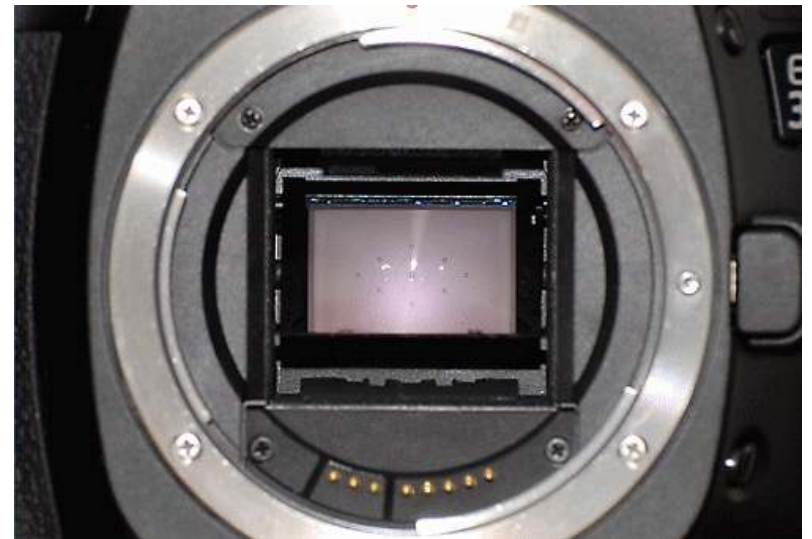
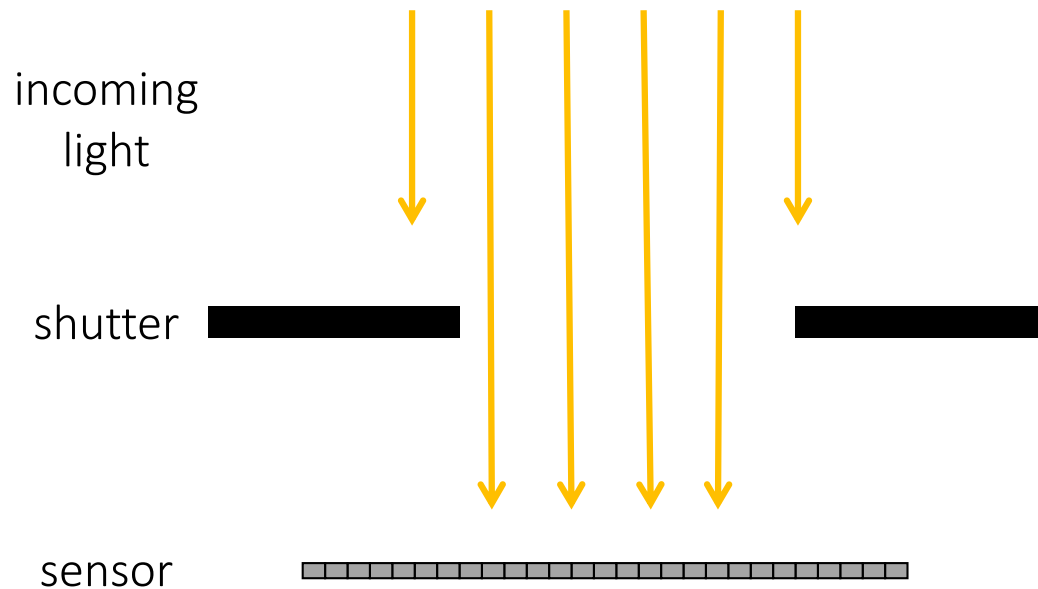
Controls the **length of time** that shutter remains open.



closed shutter

Shutter speed

Controls the **length of time** that shutter remains open.



open shutter

Shutter speed



Shutter Speed	Typical Examples
1 - 30+ seconds	Specialty night and low-light photos on a tripod
2 - 1/2 second	To add a silky look to flowing water Landscape photos on a tripod for enhanced depth of field
1/2 to 1/30 second	To add motion blur to the background of a moving subject Carefully taken hand-held photos with stabilization
1/50 - 1/100 second	Typical hand-held photos without substantial zoom
1/250 - 1/500 second	To freeze everyday sports/action subject movement Hand-held photos with substantial zoom (telephoto lens)
1/1000 - 1/4000 second	To freeze extremely fast, up-close subject motion



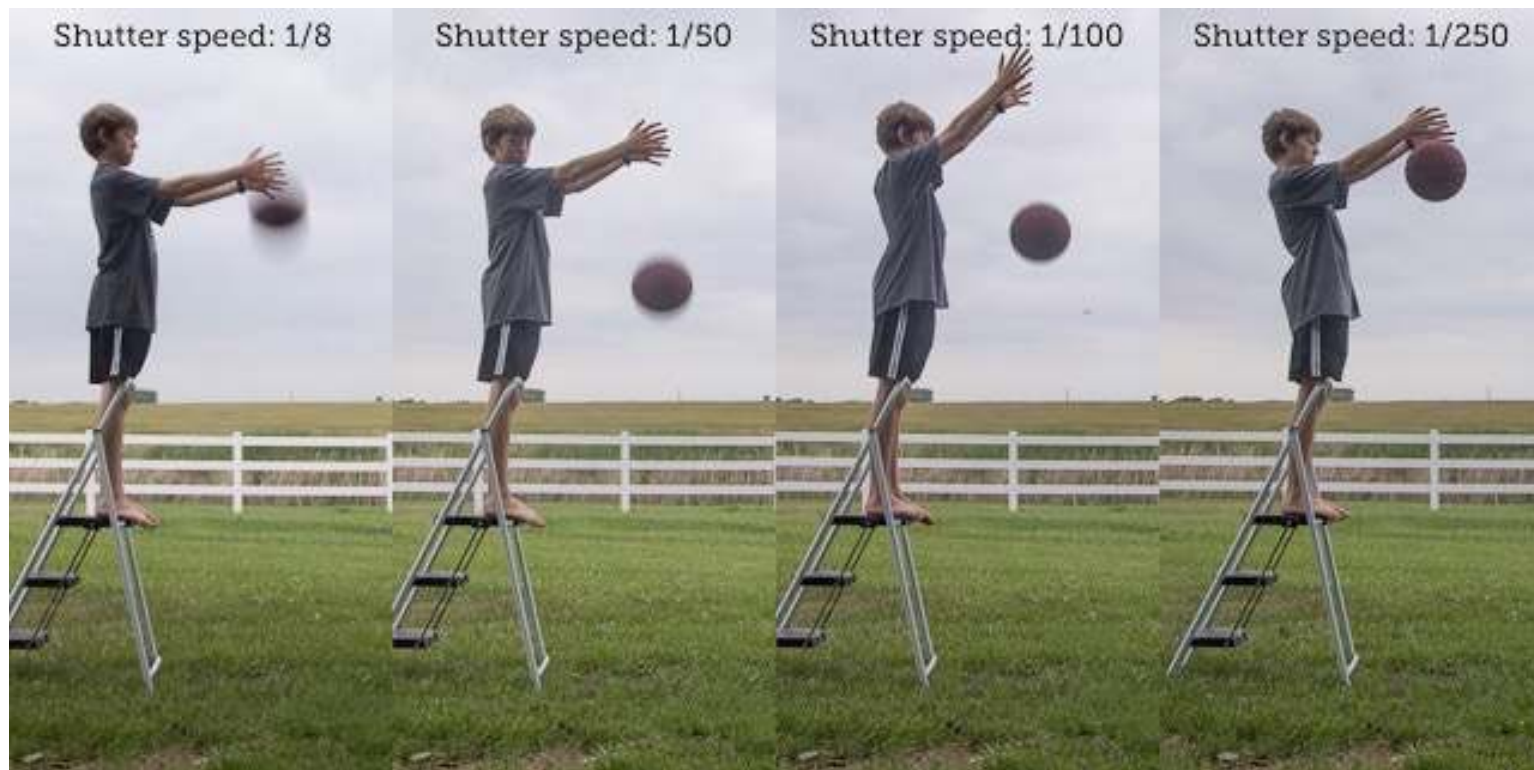
Side-effects of shutter speed

What happens to the image as we increase exposure time?

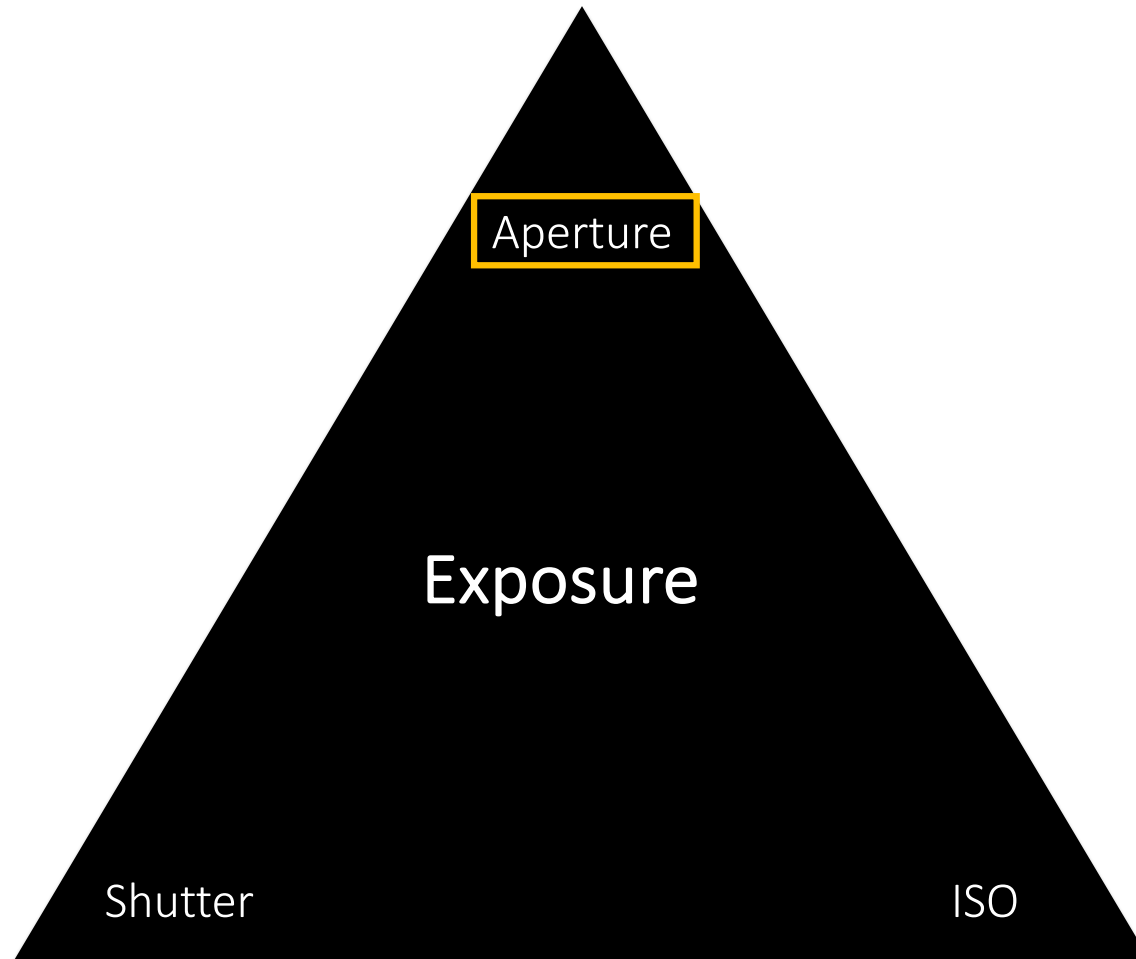


Side-effects of shutter speed

Moving scene elements appear blurry.



Exposure controls brightness of image

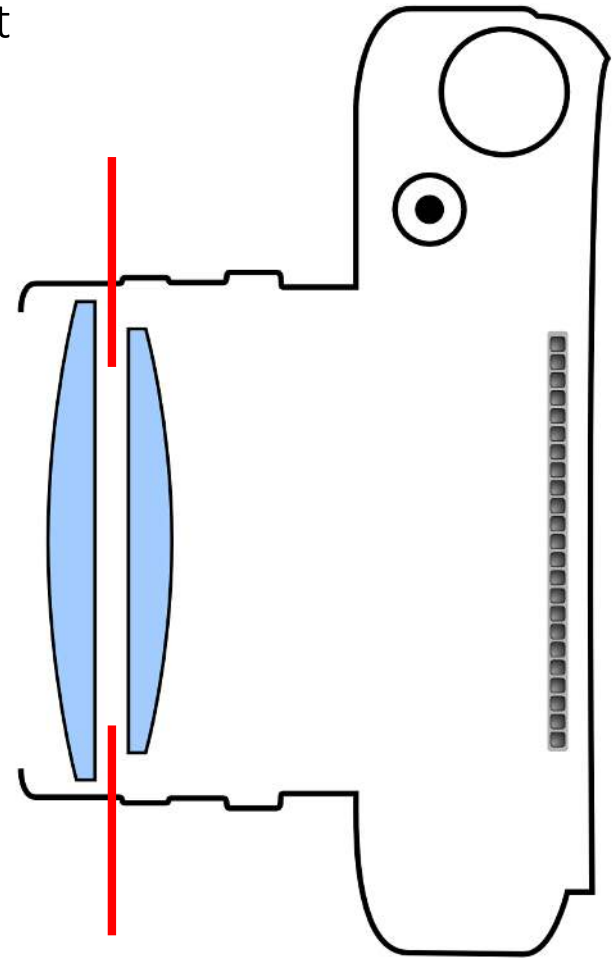


Aperture

controls amount of light



Lots of light reaches the sensor

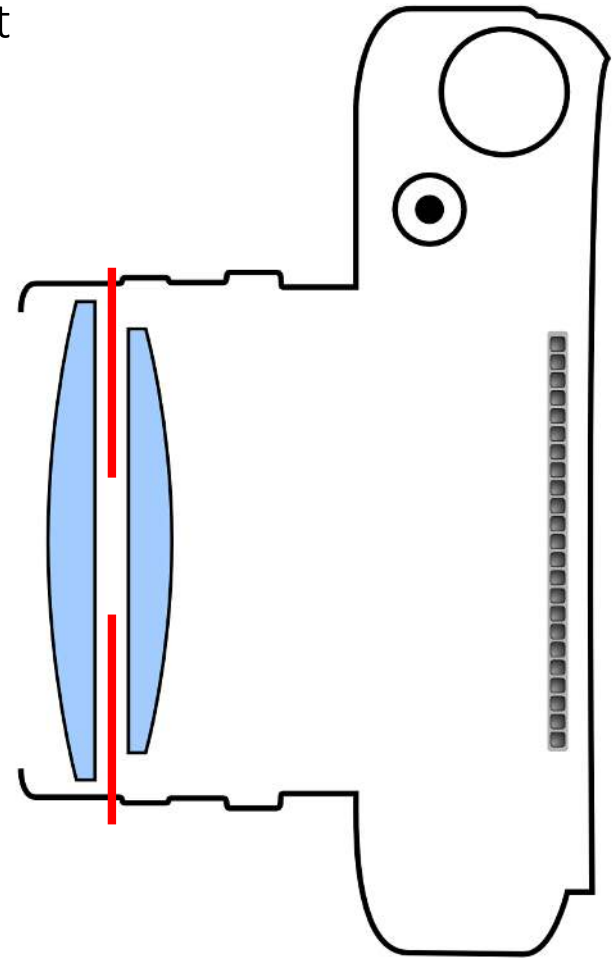


Aperture

controls amount of light



Only a little light reaches the sensor



Aperture size

Amount of irradiance is proportional to the square of the aperture diameter
AND inversely proportional to the distance to the sensor

$$f \text{ Number} = \frac{\text{Focal Length}}{\text{Aperture Diameter}}$$

Aperture size

Amount of irradiance is proposal to the square of the aperture diameter
AND inversely proposal to the distance to the sensor

$$f \text{ Number} = \frac{\text{Focal Length}}{\text{Aperture Diameter}}$$

What happens to the f number as you close the aperture?  larger  smaller



Aperture size

$$\text{f Number} = \frac{\text{Focal Length}}{\text{Aperture Diameter}}$$

What happens to the f number as you close the aperture?  larger



f/1.8



f/2.8



f/4



f/5.6



f/8

Aperture size

$$\text{f Number} = \frac{\text{Focal Length}}{\text{Aperture Diameter}}$$

What is the disadvantage of using a smaller f number (larger aperture)?



f/1.8



f/2.8



f/4

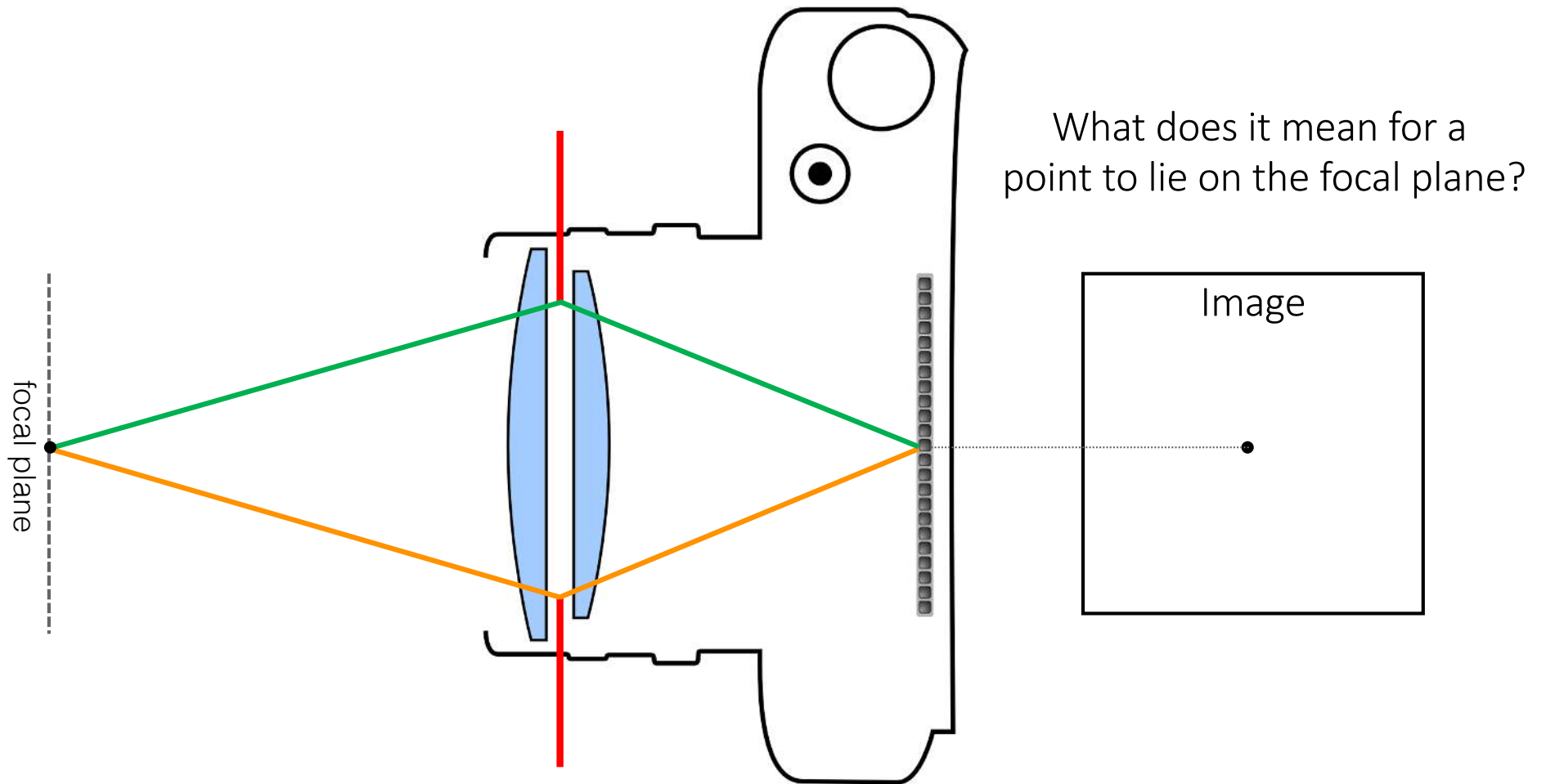


f/5.6

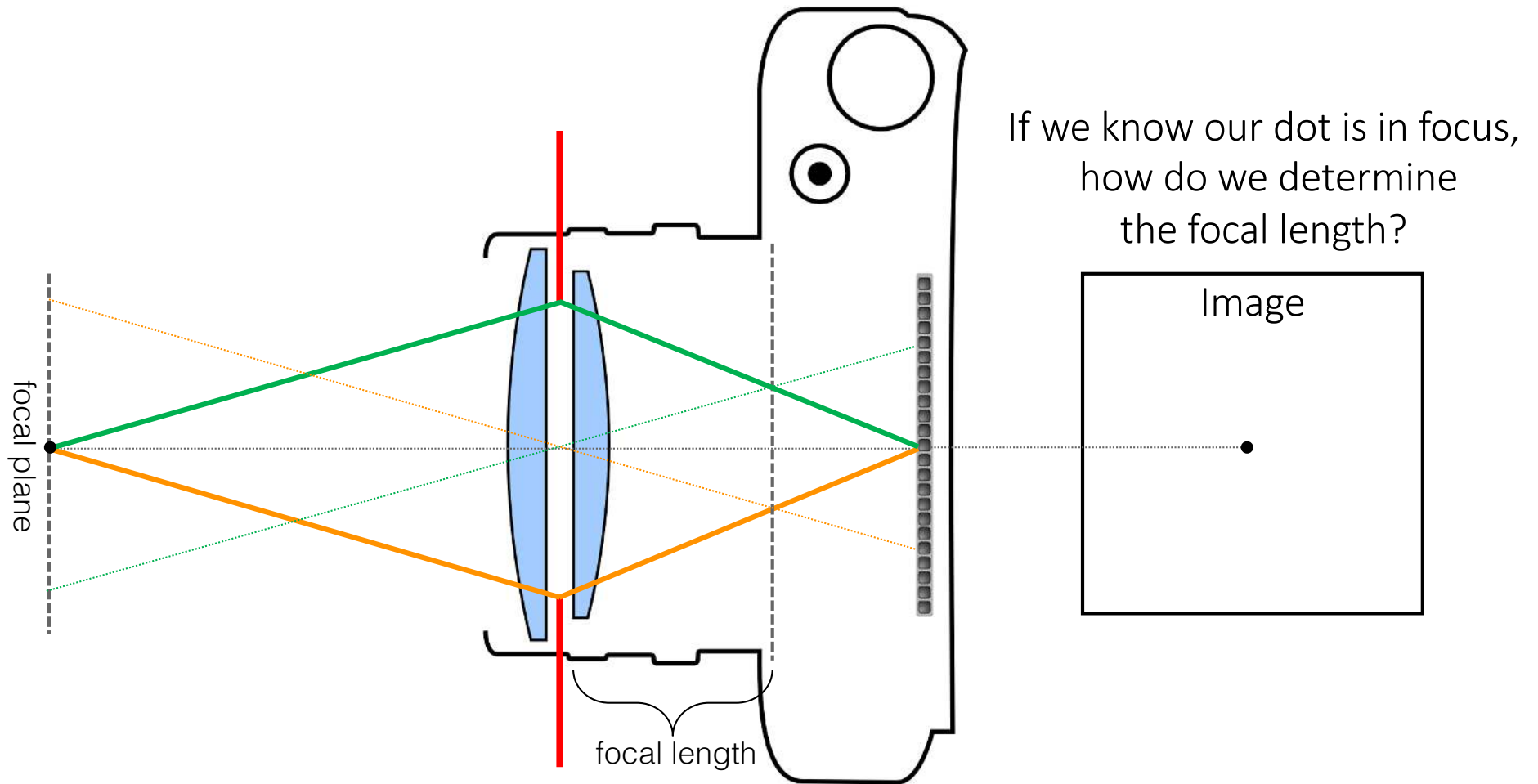


f/8

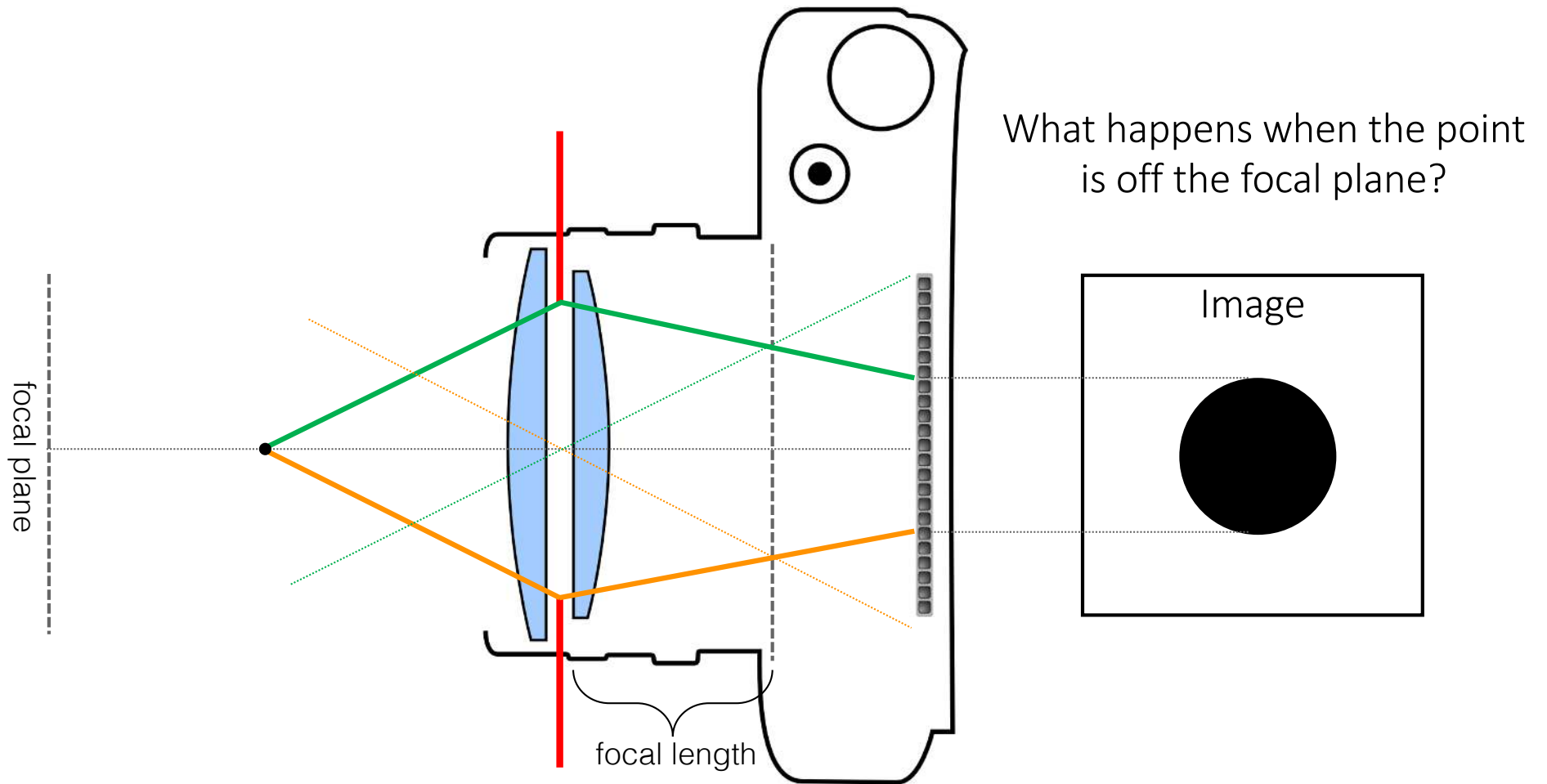
Circle of confusion



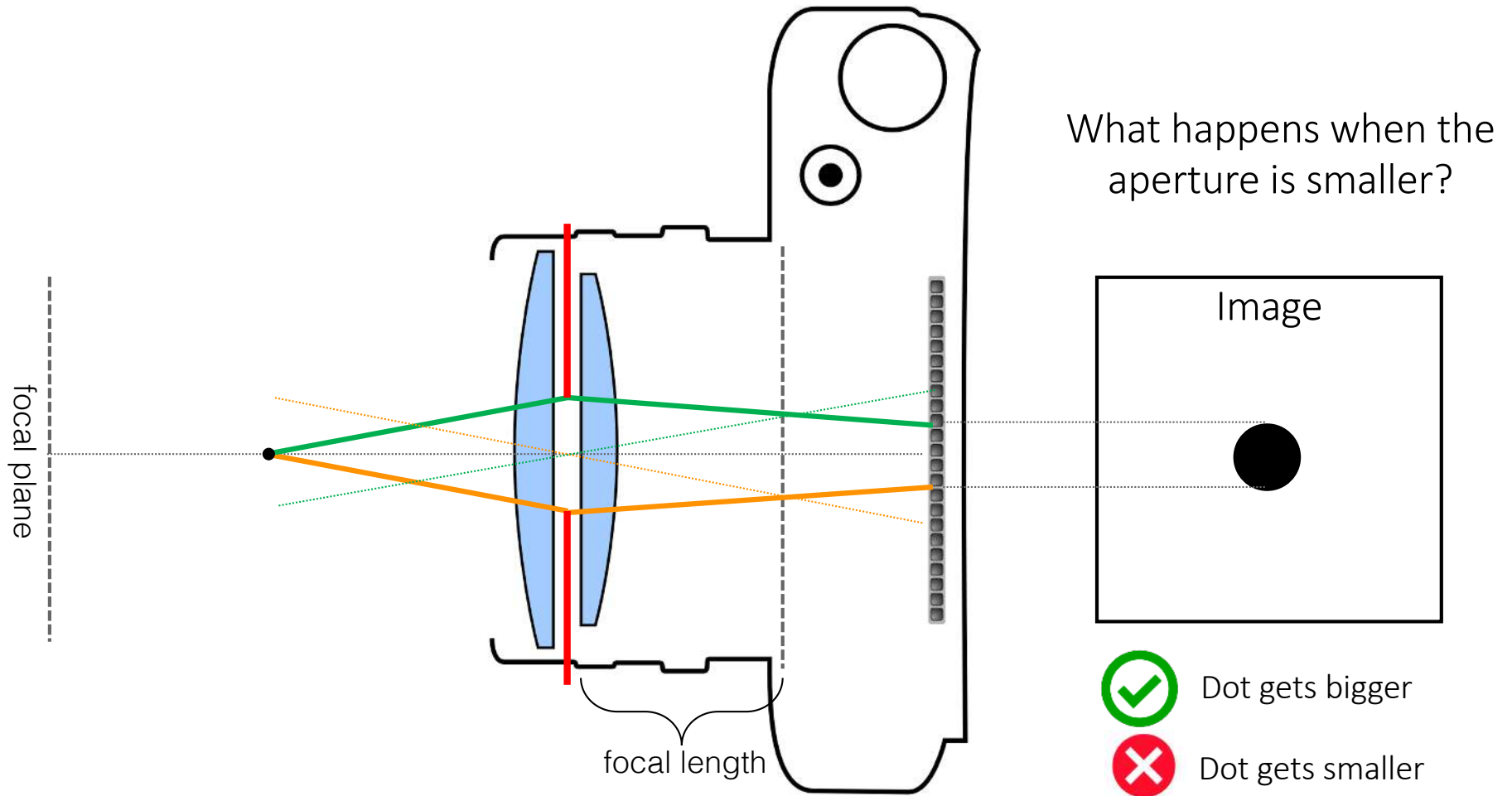
Circle of confusion



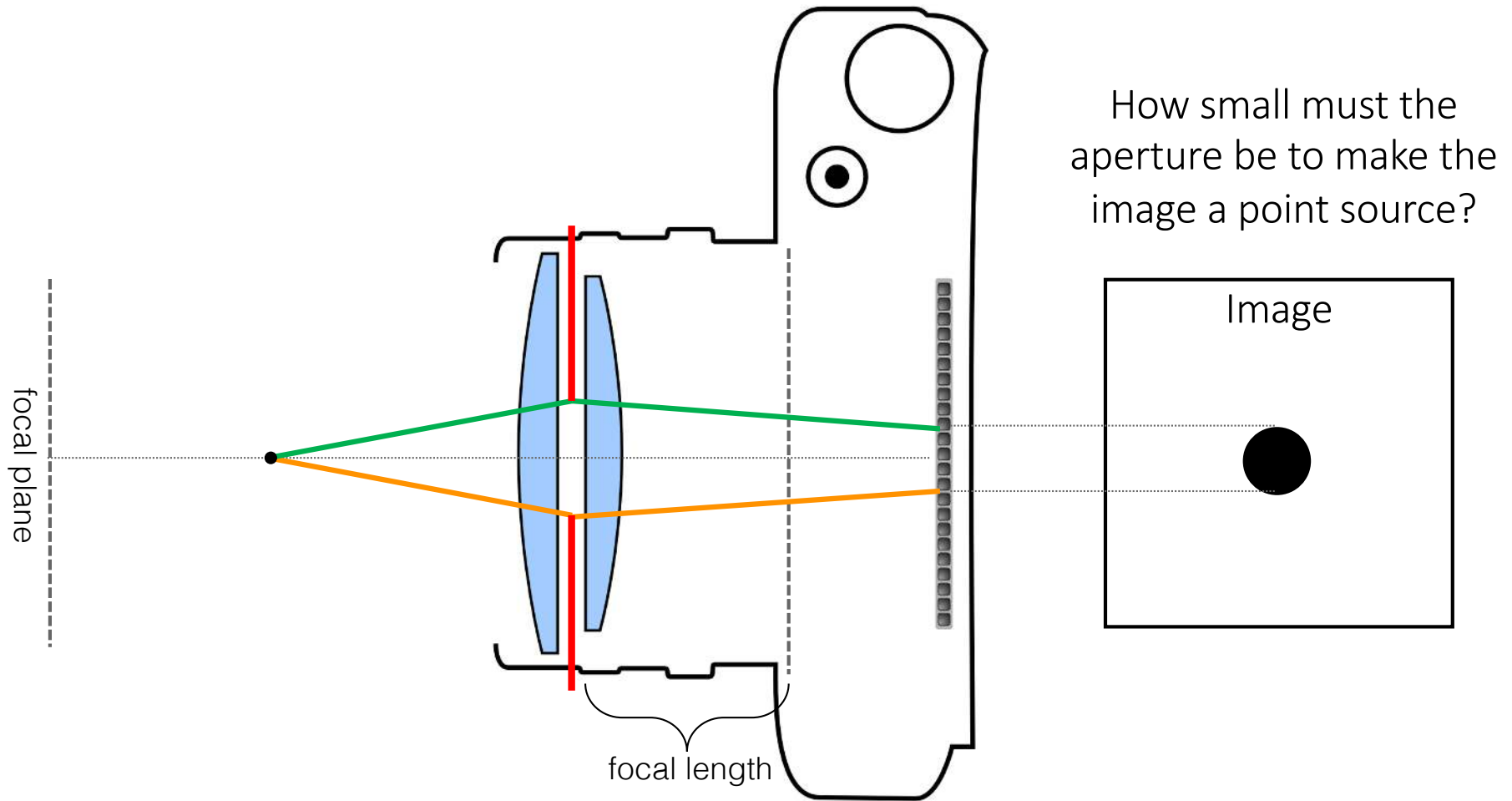
Circle of confusion



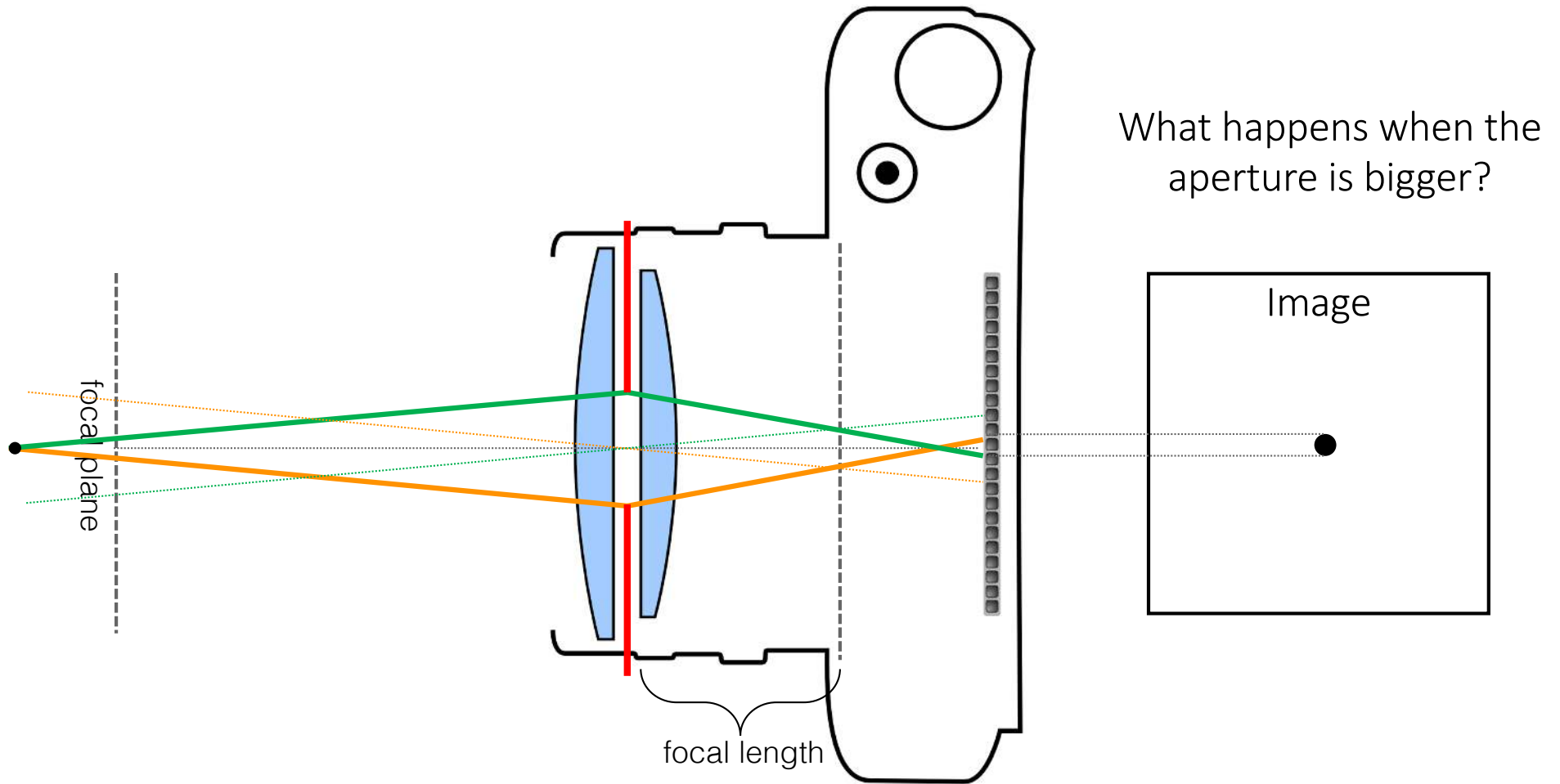
Circle of confusion



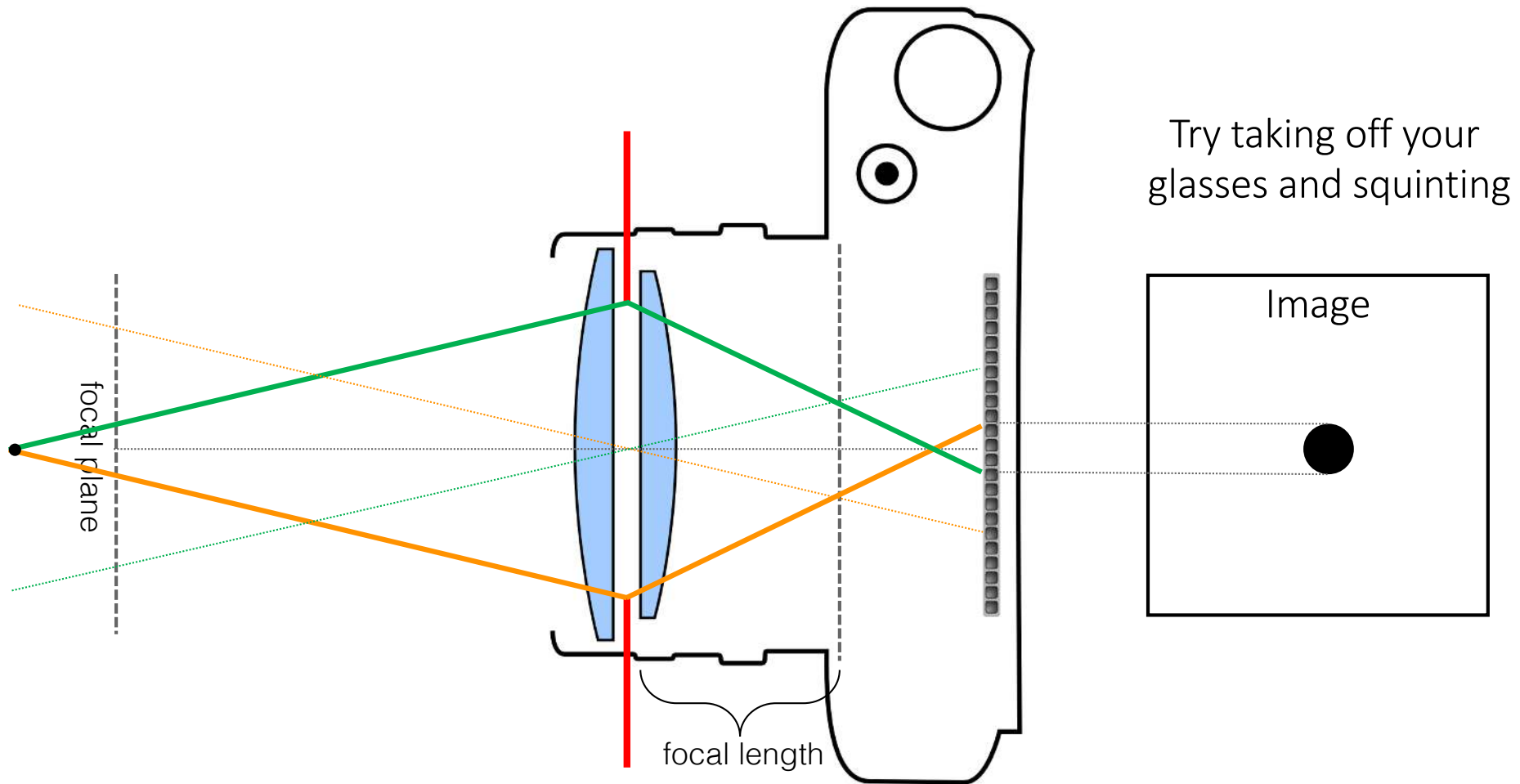
Circle of confusion



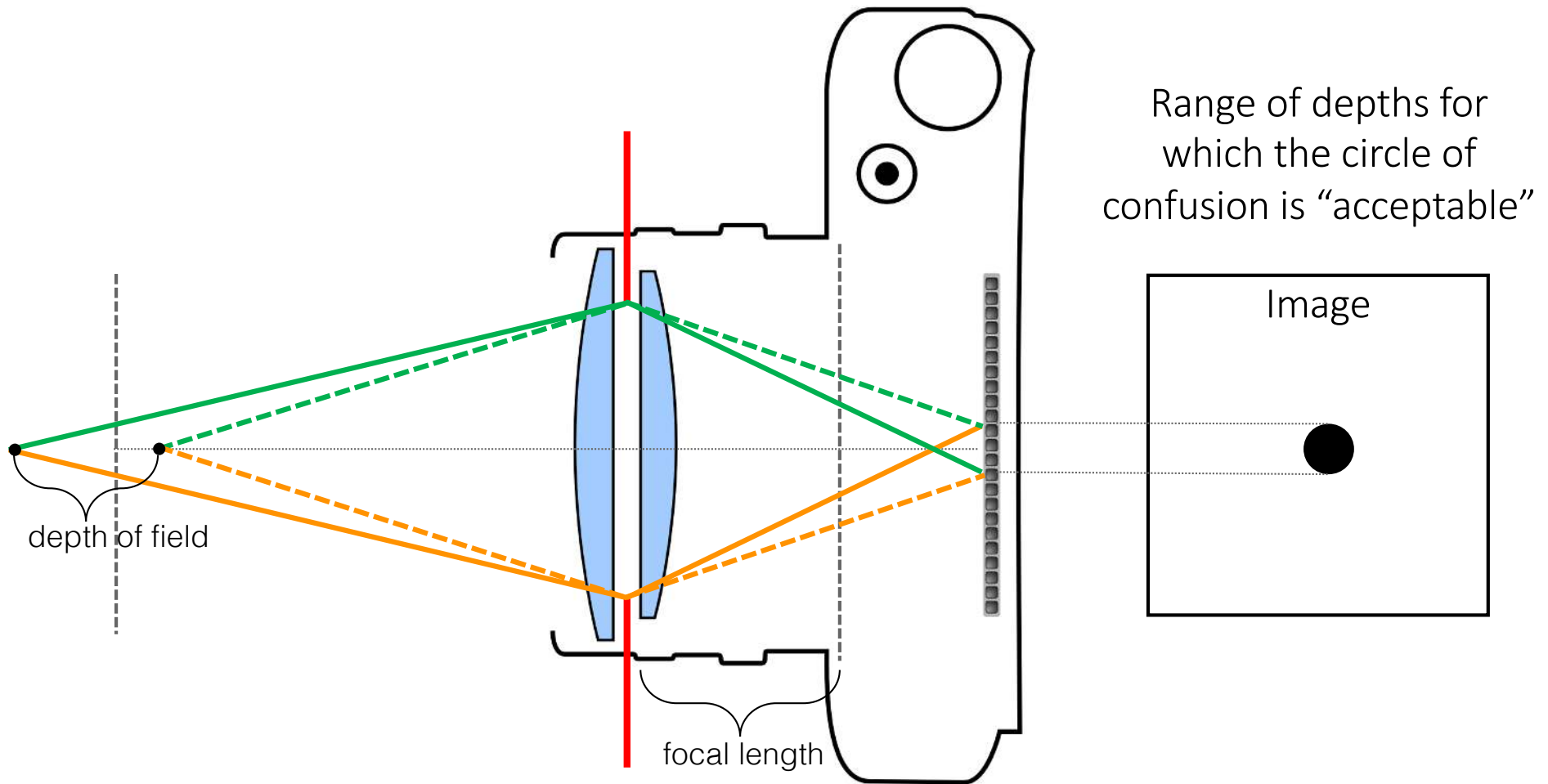
Circle of confusion



Circle of confusion



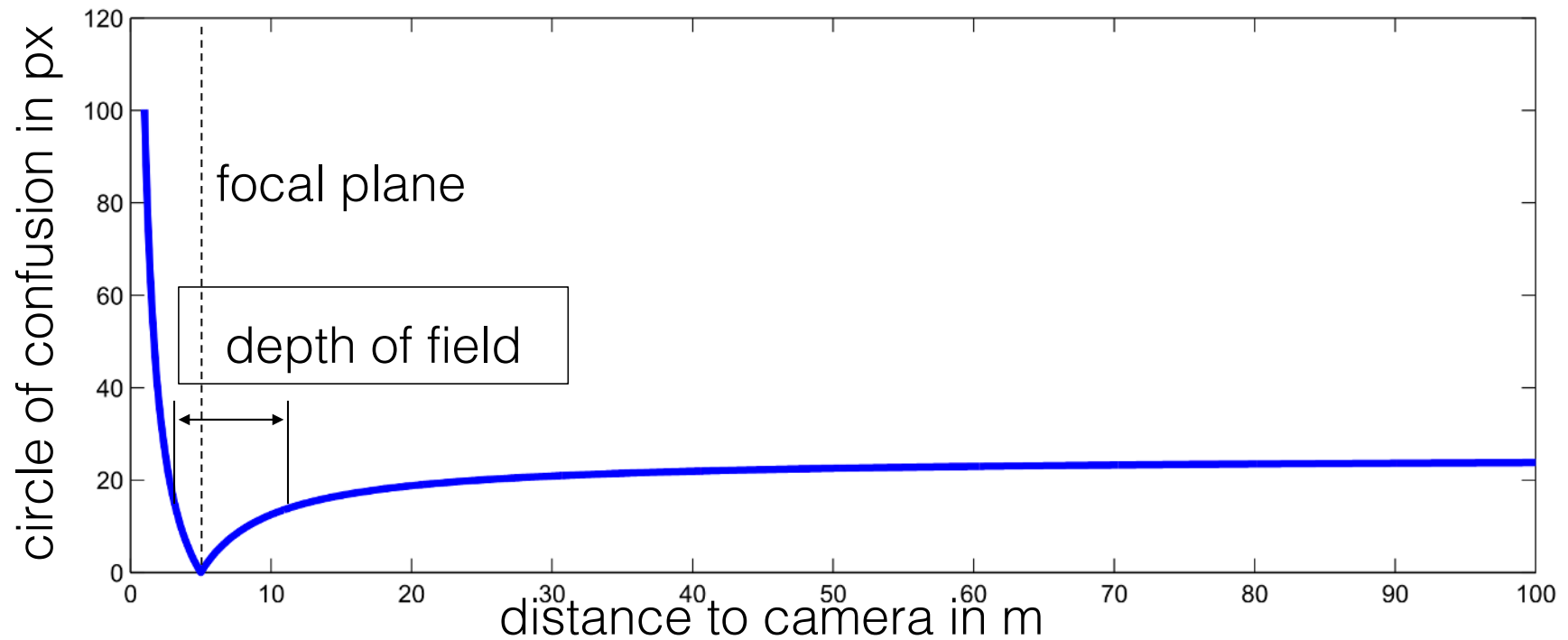
Depth of field



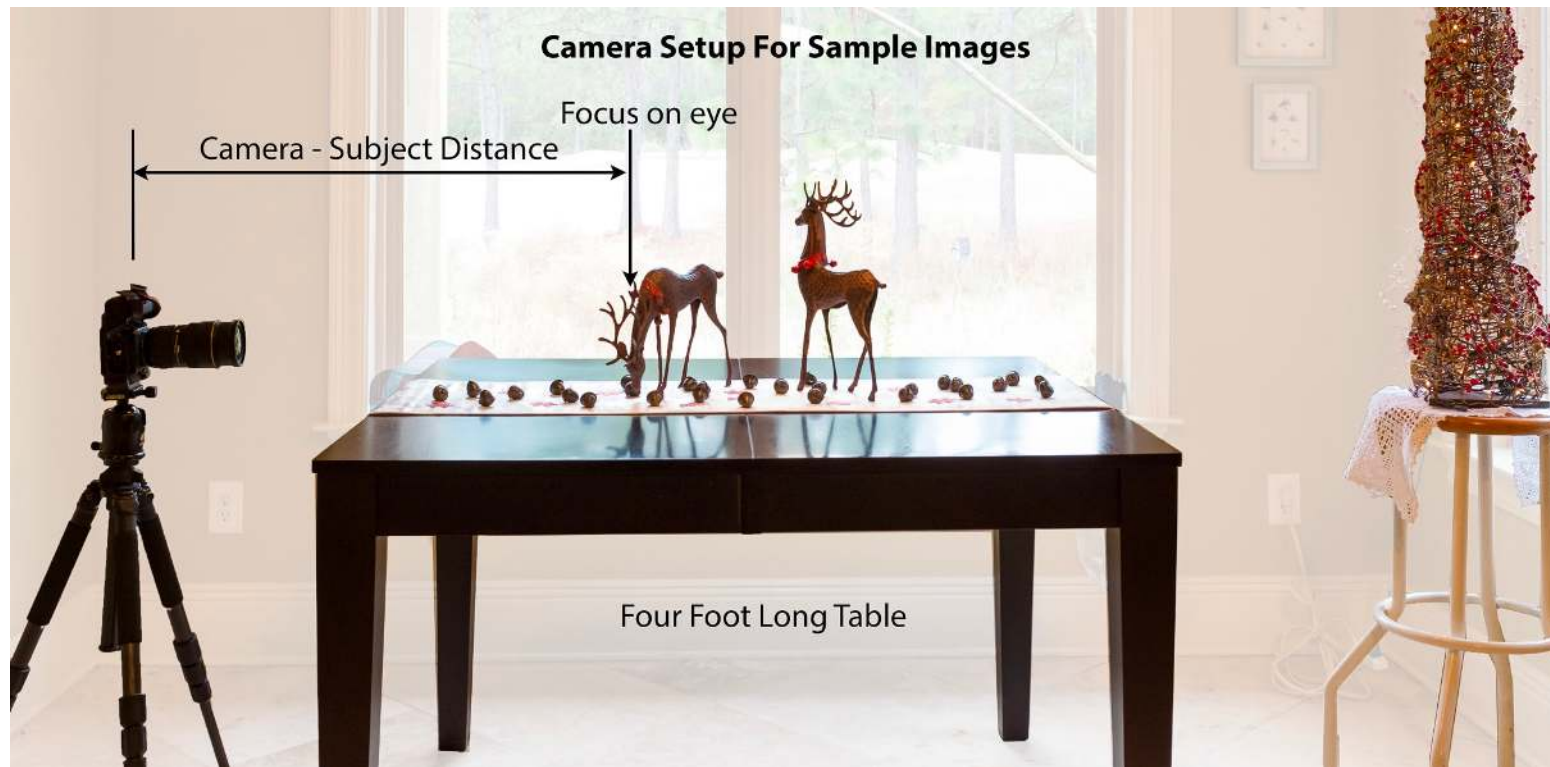
Depth of field

Canon 5D Mark III:

$f=50\text{mm}$, $f/2.8$ ($N=2.8$), focused at 5m , pixel size= $7.5\mu\text{m}$



Depth of field



<https://photographylife.com/what-is-depth-of-field>

Depth of field

A larger f number (smaller aperture) → larger depth of field



f/3.2, dof = 0.024m

f/5.6, dof = 0.043m

f/11, dof = 0.086m

f/22, dof = 0.172m

Focal length 105mm, Focus Distance 1.5m

<https://photographylife.com/what-is-depth-of-field>

Depth of field

Which picture was taken with a smaller f number?



Depth of field

Sharp depth of field (“bokeh”) is often desirable



How can you simulate bokeh?

Infer per-pixel depth, then blur with depth-dependent kernel.

- Example: Google camera “lens blur” feature



Barron et al., “Fast Bilateral-Space Stereo for Synthetic Defocus,” CVPR 2015

Aperture size and Exposure



←
more light
...means more exposure
...means faster shutter speed

Fastest lenses commercially available

In consumer photography, fastest lenses are f/0.9 – f/0.95.

$$f \text{ Number} = \frac{\text{Focal Length}}{\text{Aperture Diameter}}$$



Leica Noctilux 50mm f/0.95 (Price tag: > \$10,000)

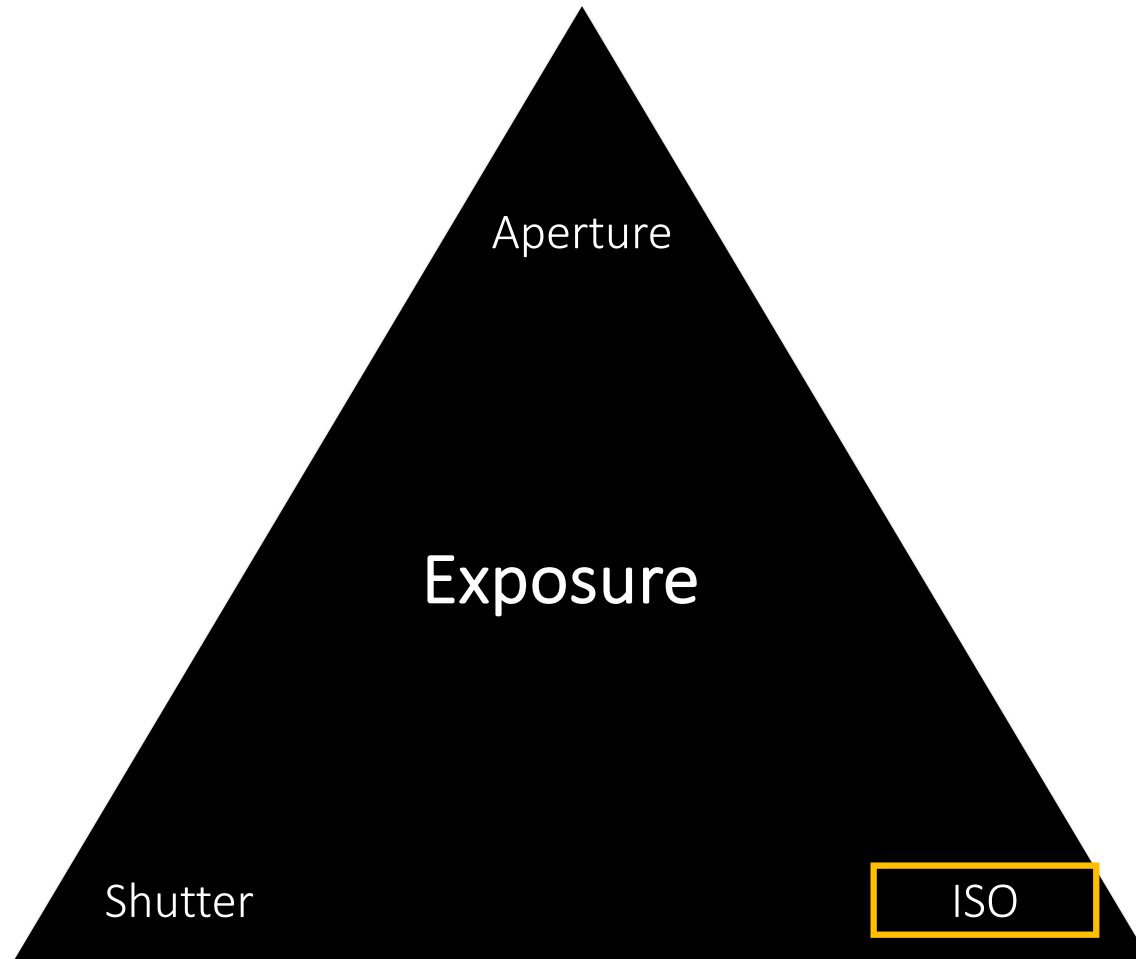
Fastest lens ever made?

Zeiss 50 mm f / 0.7 Planar lens



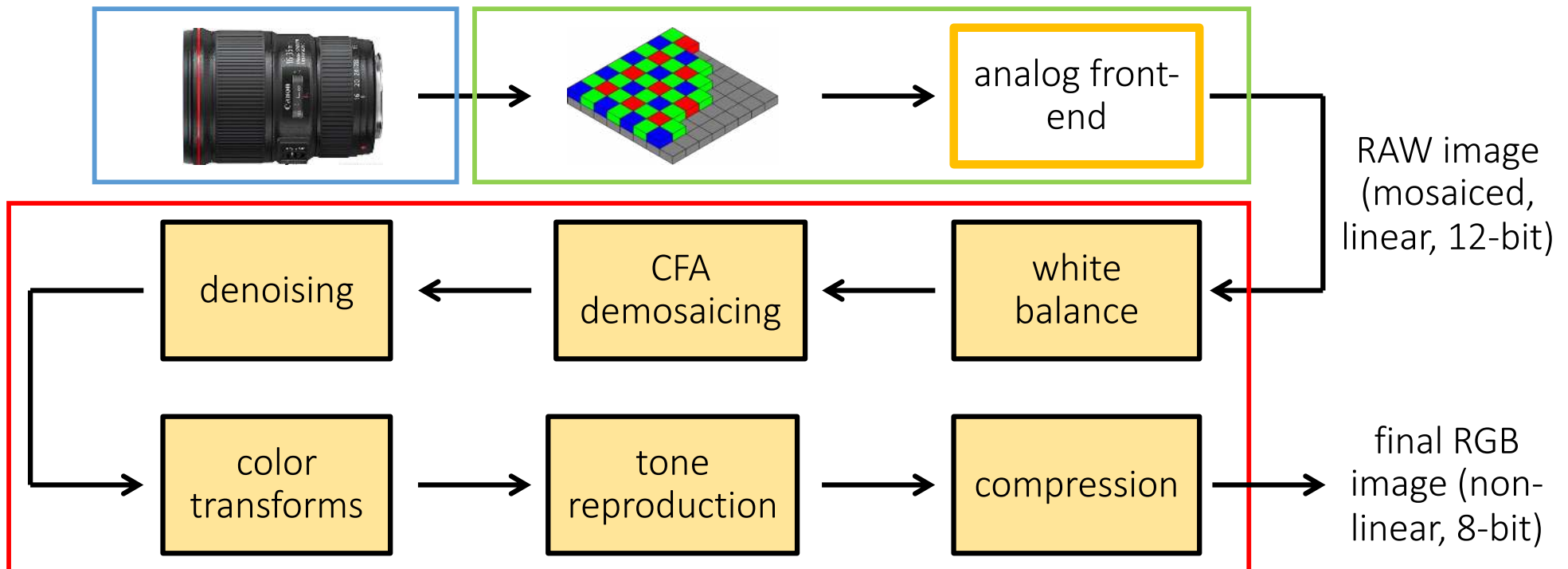
- Originally developed for NASA's Apollo missions.
- Stanley Kubrick somehow got to use the lens to shoot movies under only candlelight.

Exposure controls brightness of image

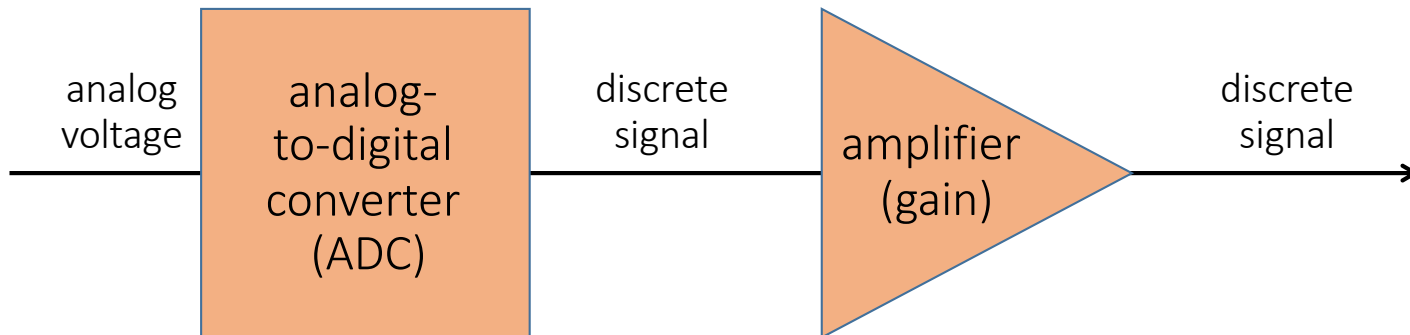
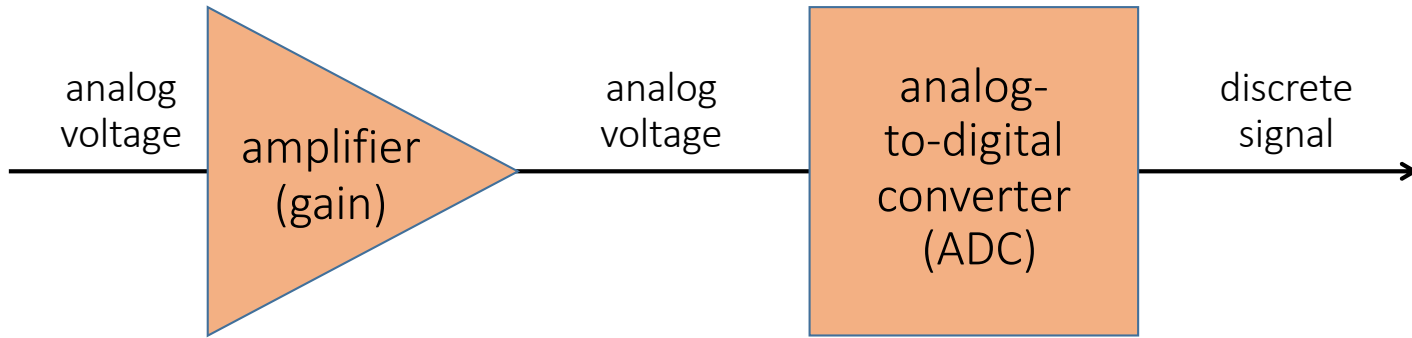


The (in-camera) image processing pipeline

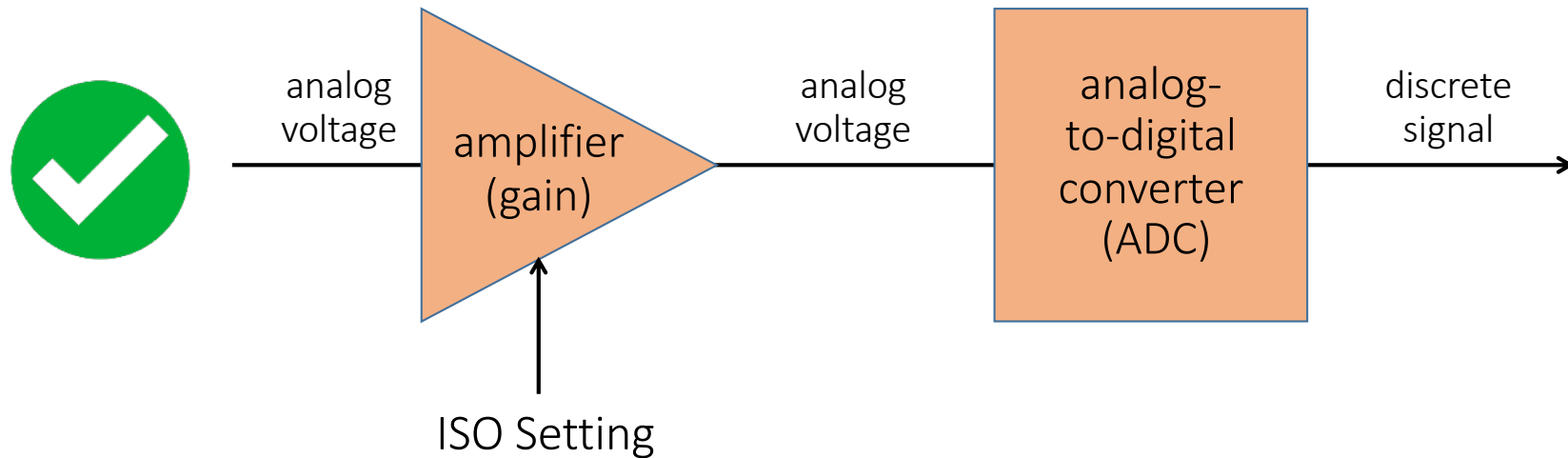
The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



Which is better?



Which is better?



ISO is not an acronym.

- It refers to the International **O**rganization for **S**tandardization.
- ISO comes from the Greek word *ἴσος*, which means *equal*.
- It is pronounced (roughly) *eye-zo*

Side-effects of increasing ISO

Image becomes very grainy because noise is amplified.



ISO 80

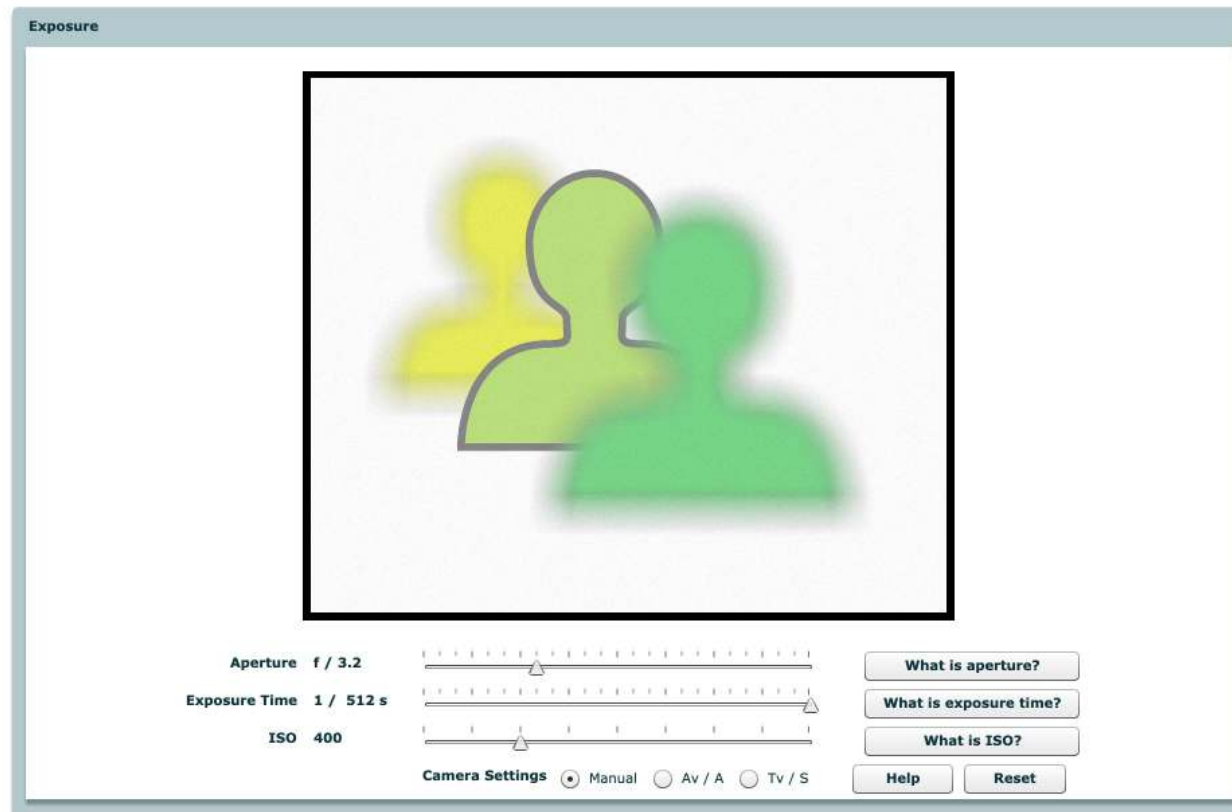


ISO 800



ISO 1600

<http://graphics.stanford.edu/courses/cs178/applets/exposure.html>



Camera modes

Aperture priority (“A”): you set aperture, camera sets everything else.

- Pros: Direct depth of field control.
- Cons: Can require impossible shutter speed (e.g. with f/1.4 for a bright scene).

Shutter speed priority (“S”): you set shutter speed, camera sets everything else.

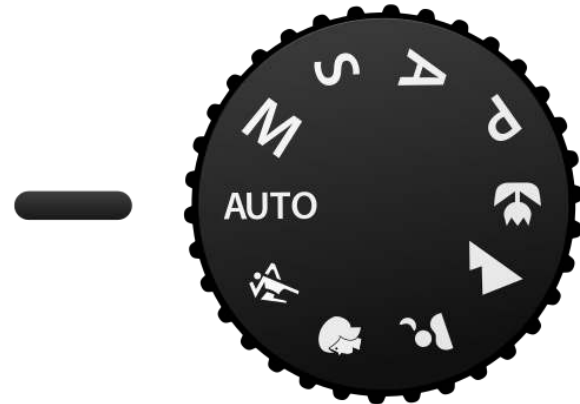
- Pros: Direct motion blur control.
- Cons: Can require impossible aperture (e.g. when requesting a 1/1000 speed for a dark scene)

Automatic (“AUTO”): camera sets everything.

- Pros: Very fast, requires no experience.
- Cons: No control.

Manual (“M”): you set everything.

- Pros: Full control.
- Cons: Very slow, requires a lot of experience.



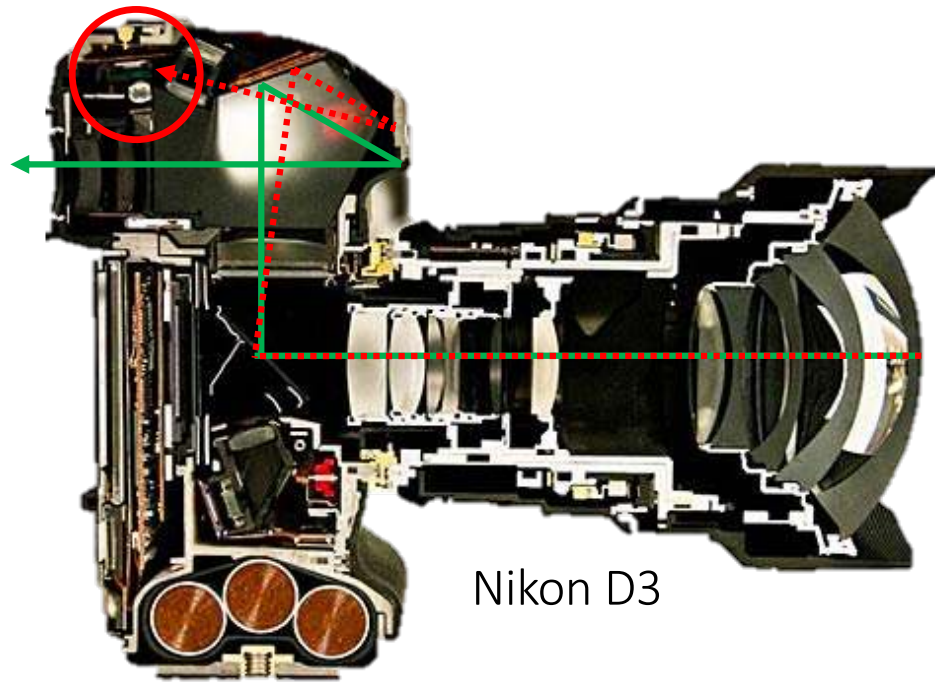
Light metering

Light metering in modern cameras

- SLR cameras use a separate low-resolution sensor, with big pixels, that is placed at the focusing screen (e.g., 20 bit pixels)
- Mirrorless cameras use measurements directly from the main sensor.



Nikon D3X



Nikon D3

Marc Levoy
Steve Stenzel

Metering challenges: low resolution

Low-resolution can make it difficult to correctly meter the scene and set exposure.

- In which of these scenes is it OK to let the brightest pixels be overexposed?



Metering challenges: low resolution

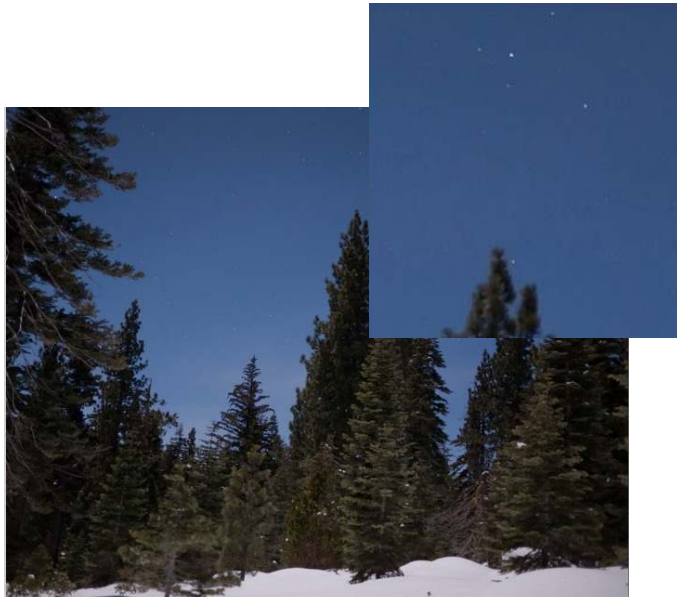
Low-resolution can make it difficult to correctly meter the scene and set exposure.

- In which of these scenes is it OK to let the brightest pixels be overexposed?



Our devices do not match the world

The world has a high dynamic range



Moonlight

8s, f/1.7, ISO 400

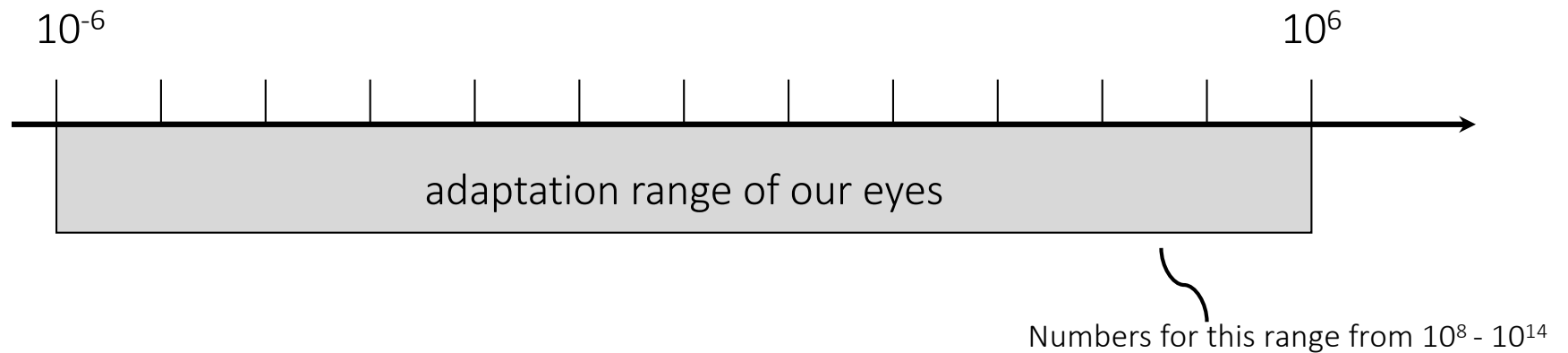


Daylight

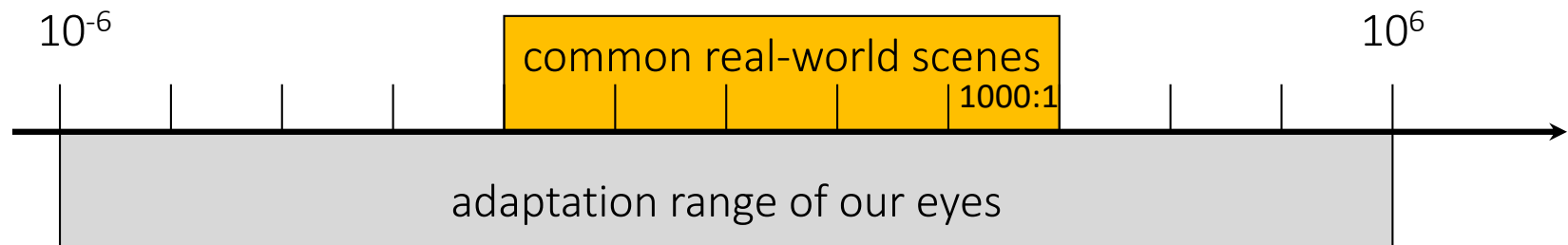
1/500s, f/6.3, ISO 100

1 : 256,000

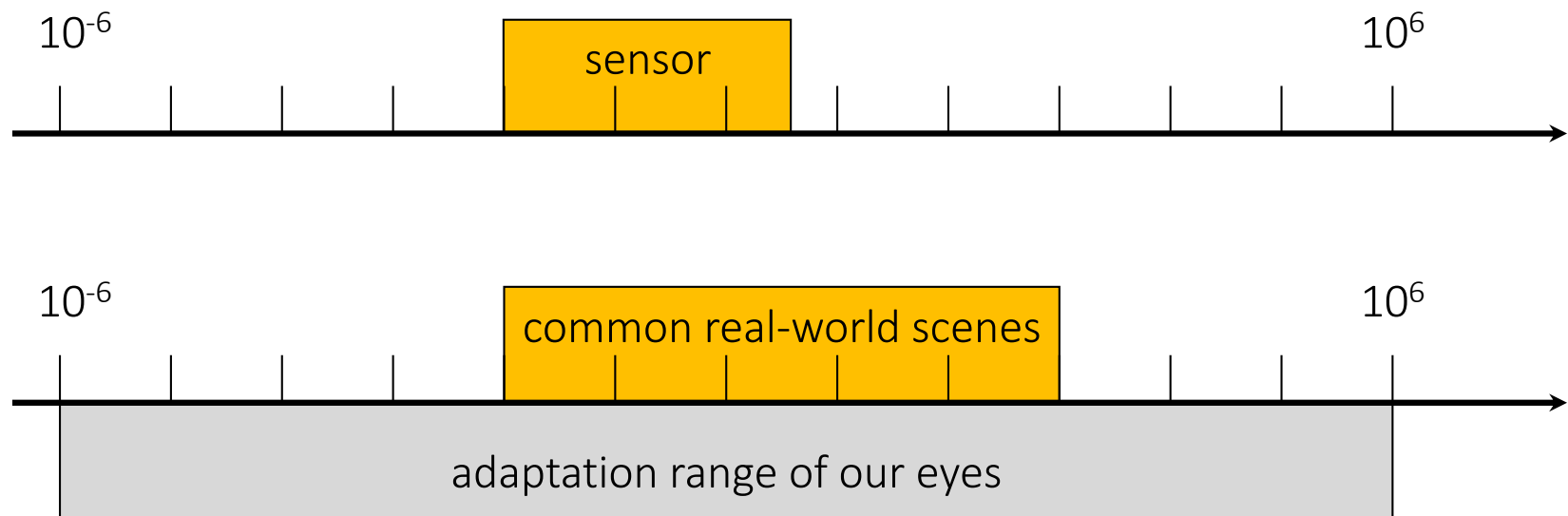
The world has a high dynamic range



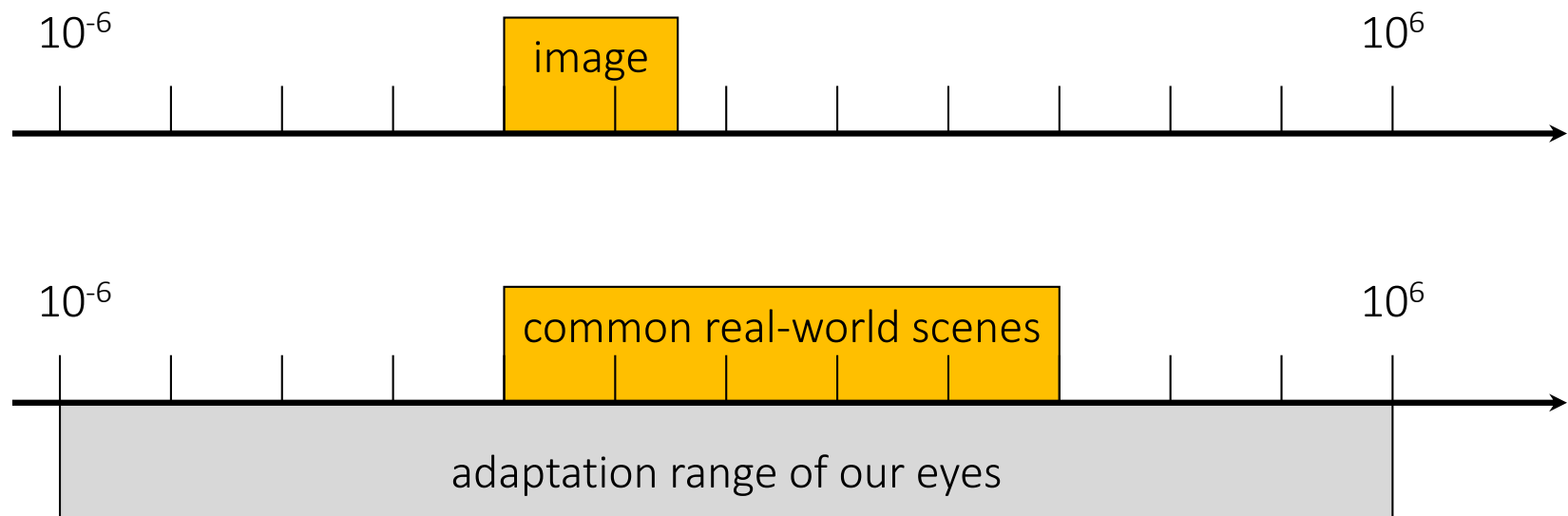
The world has a high dynamic range



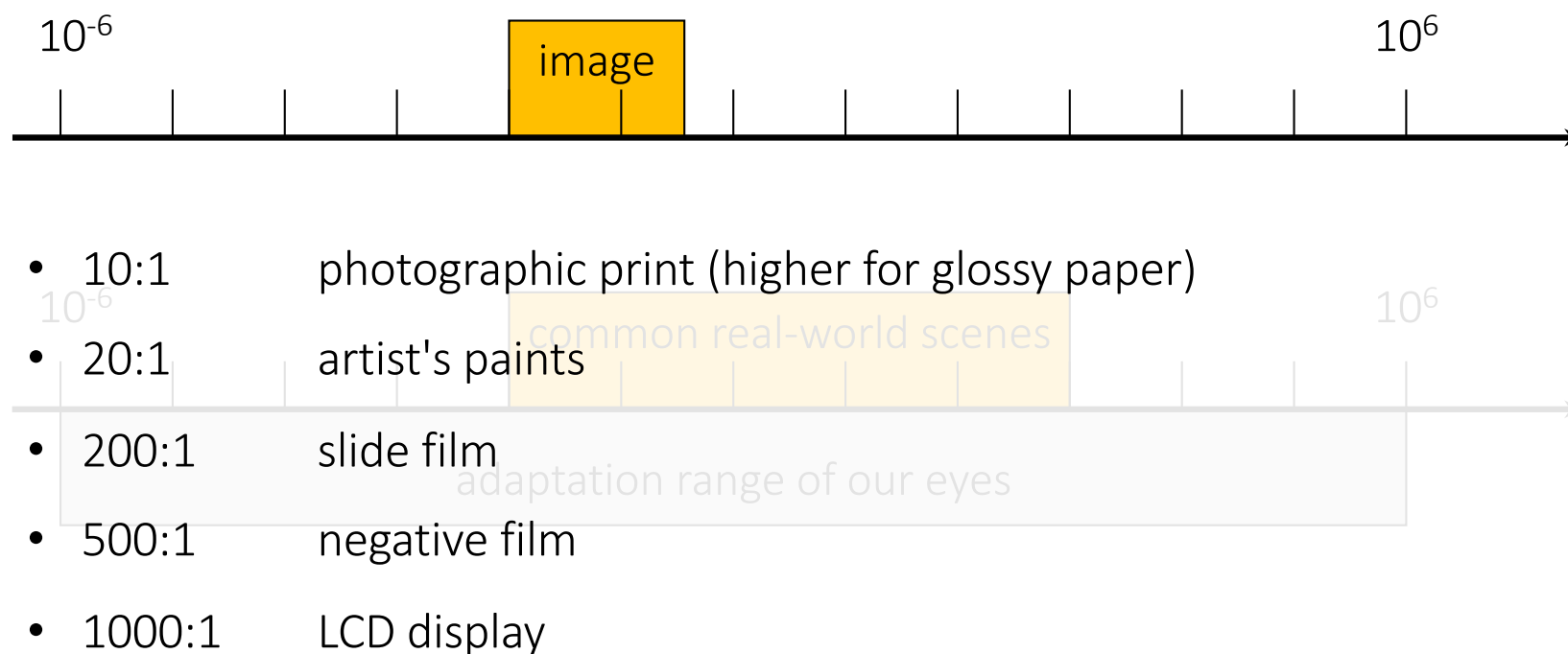
(Digital) sensors also have a low dynamic range



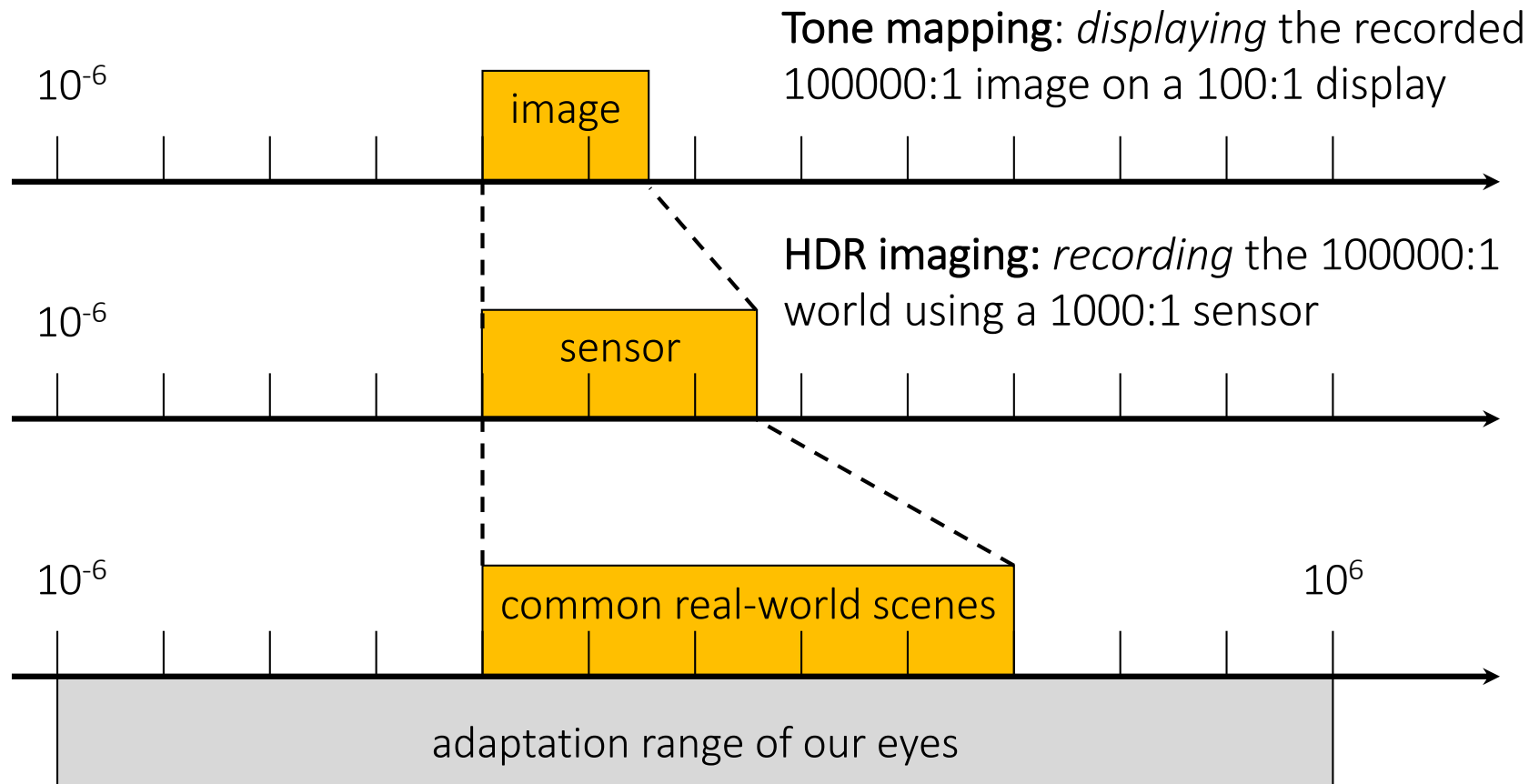
(Digital) images have an even lower dynamic range



(Digital) images have an even lower dynamic range

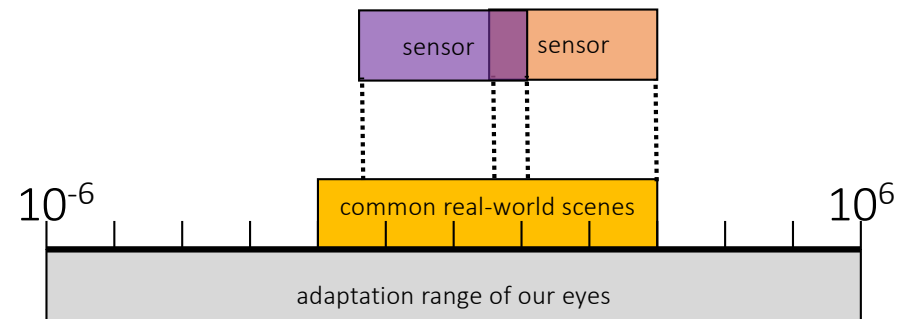


HDR imaging & Tone mapping

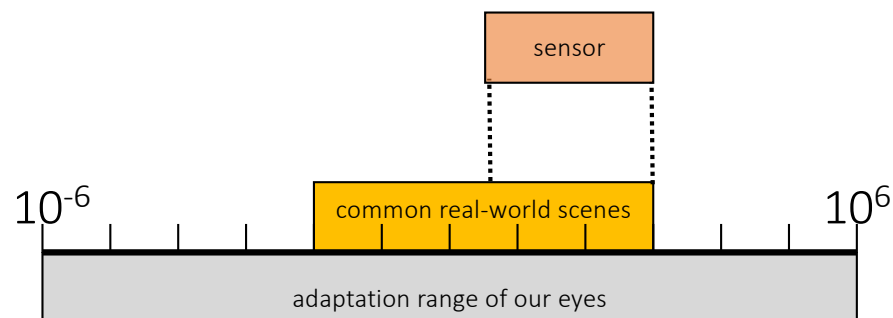


High dynamic range imaging

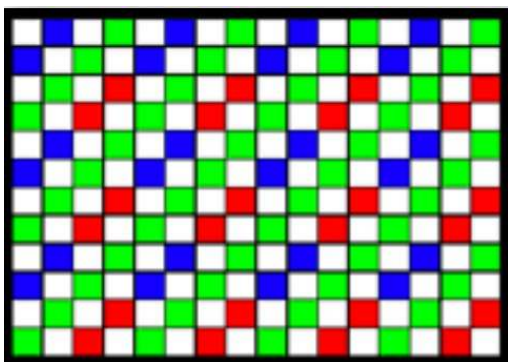
The dynamic range problem



How can we capture a high dynamic range image?

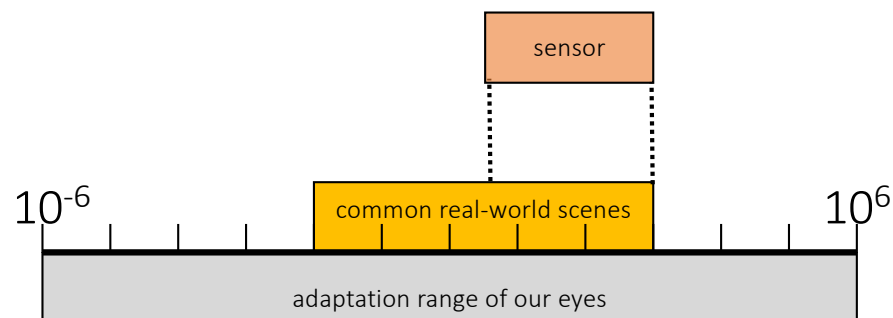


alternative color filter arrays

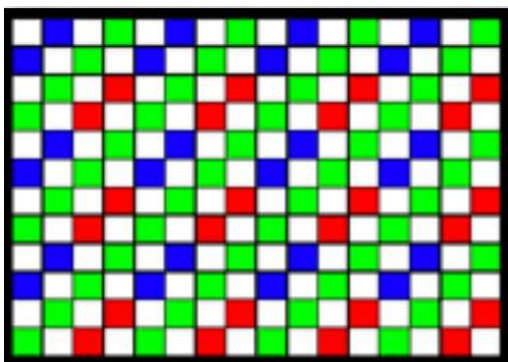


RGBC (a.k.a. RGBW)

How can we capture a high dynamic range image?

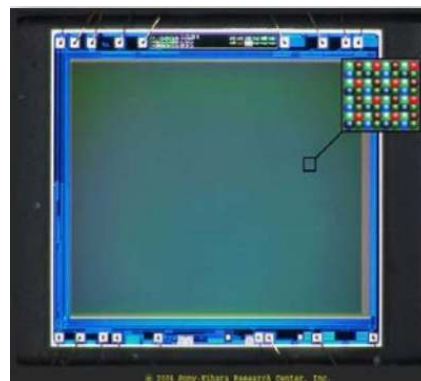


alternative color filter arrays



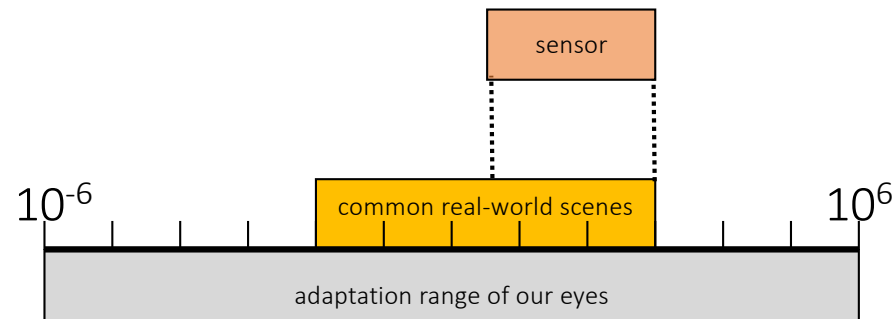
RGBC (a.k.a. RGBW)

per-pixel neutral density filters



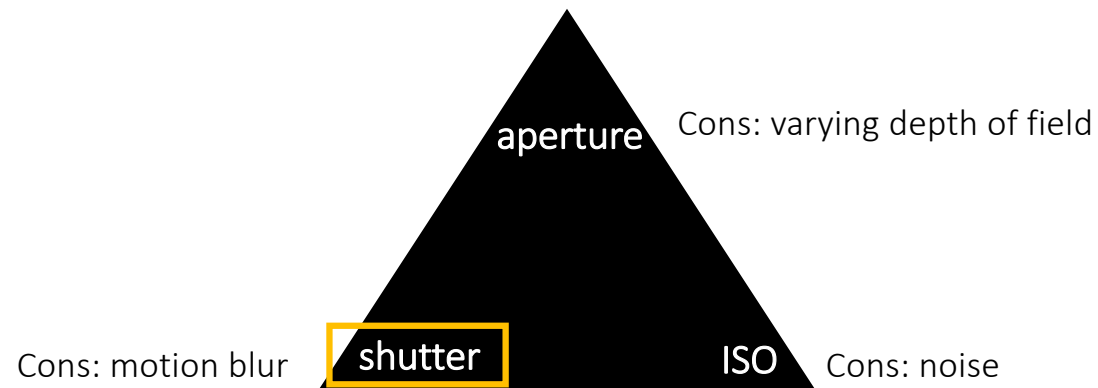
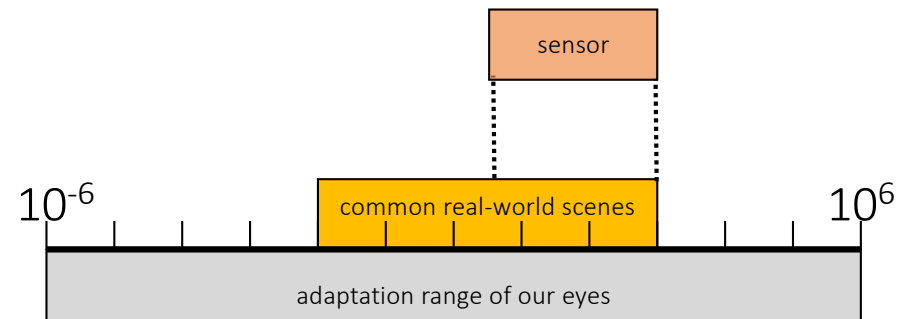
[Nayar CVPR 2000] throws away photons

How can we capture a high dynamic range image?

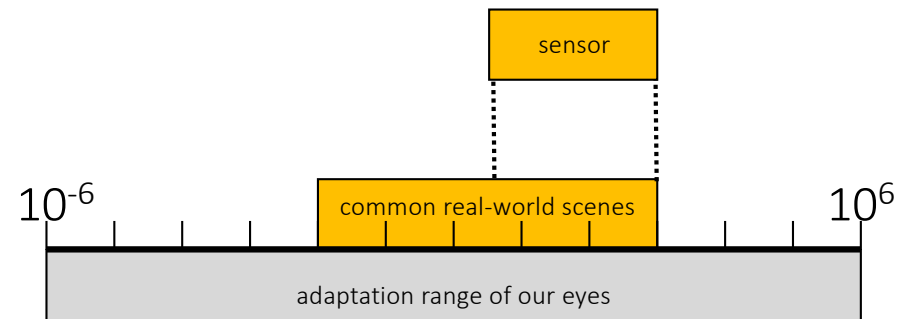


1. Exposure bracketing: capture multiple LDR images at different exposures
2. Merging: combine them into a single HDR image

How should we vary the exposure?

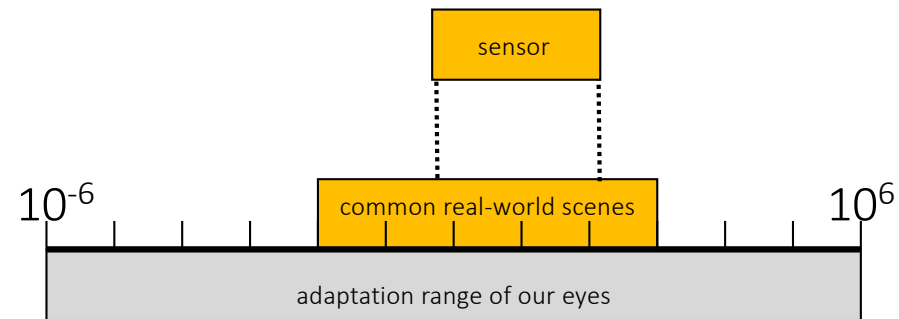


Multiple exposure bracketing



t_1

Multiple exposure bracketing

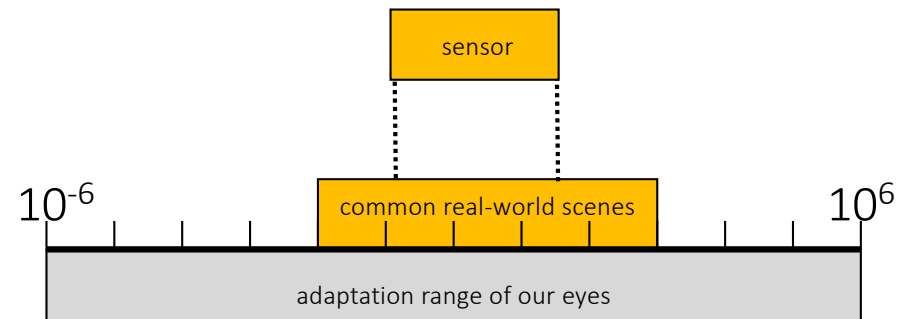


t_2



t_1

Multiple exposure bracketing



t_3

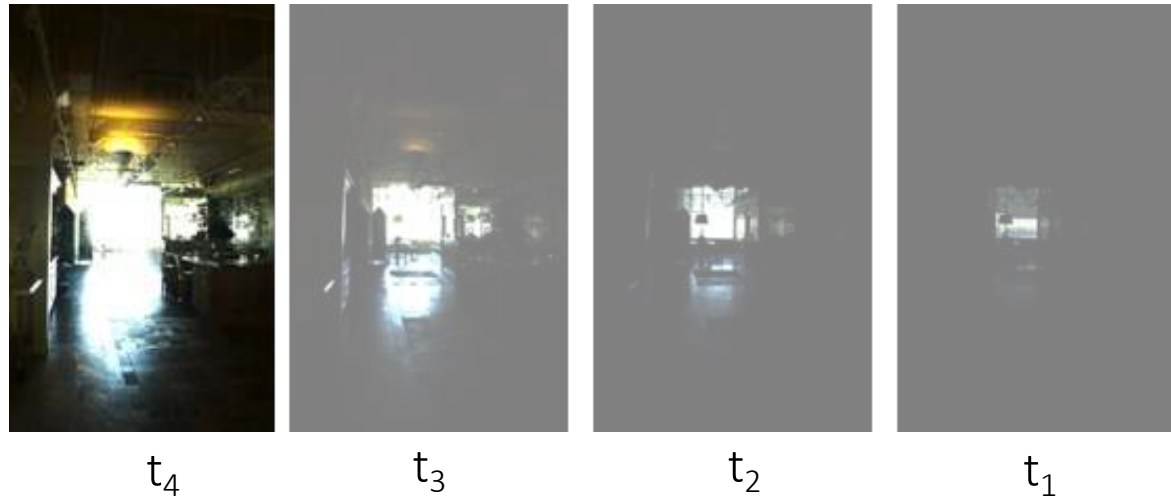
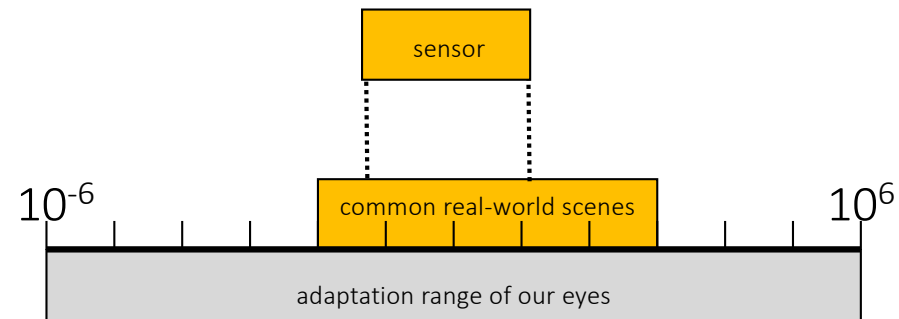


t_2

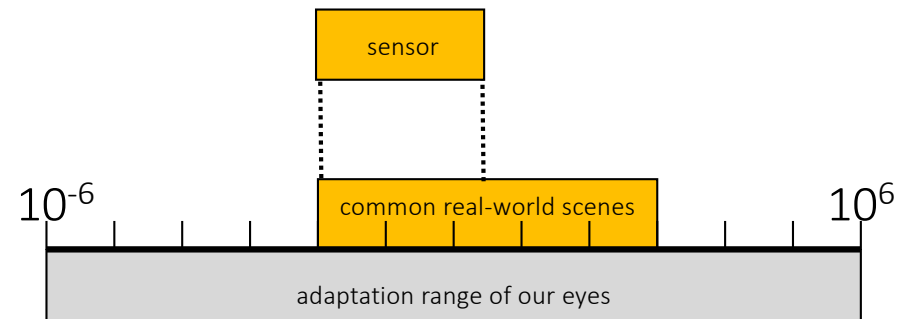


t_1

Multiple exposure bracketing



Multiple exposure bracketing



t_5



t_4



t_3

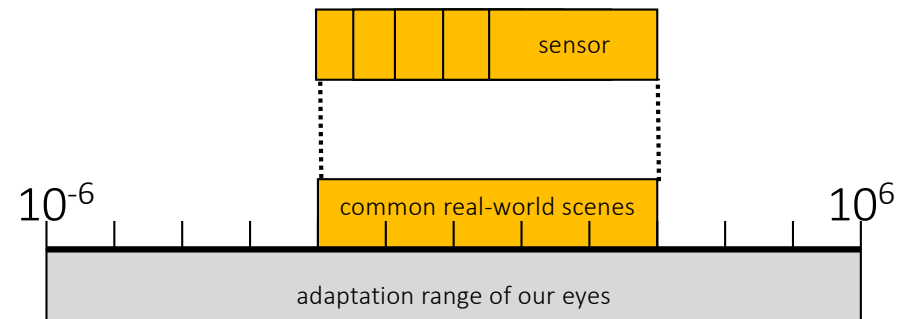


t_2



t_1

Multiple exposure bracketing



t_5



t_4



t_3

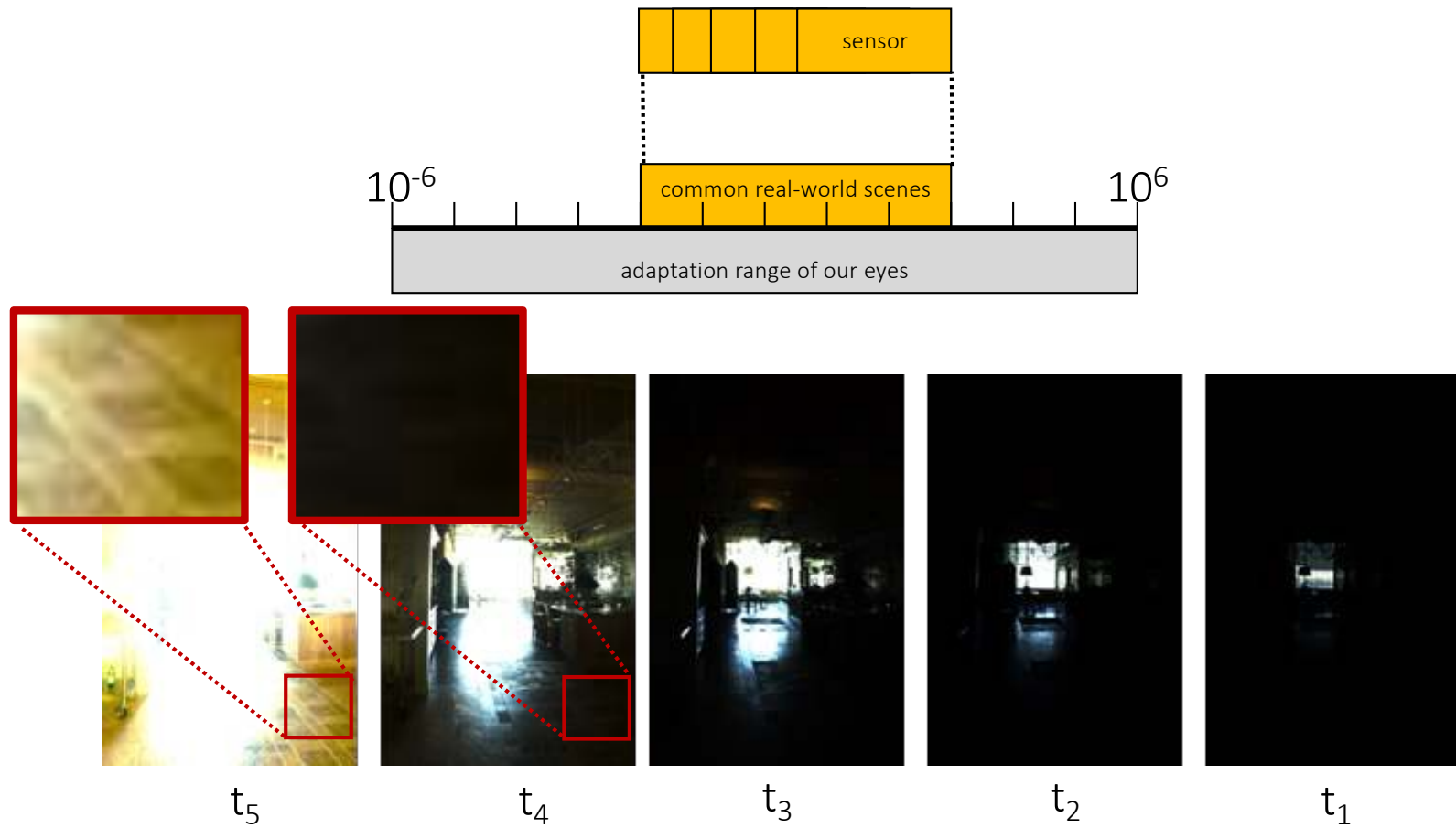


t_2

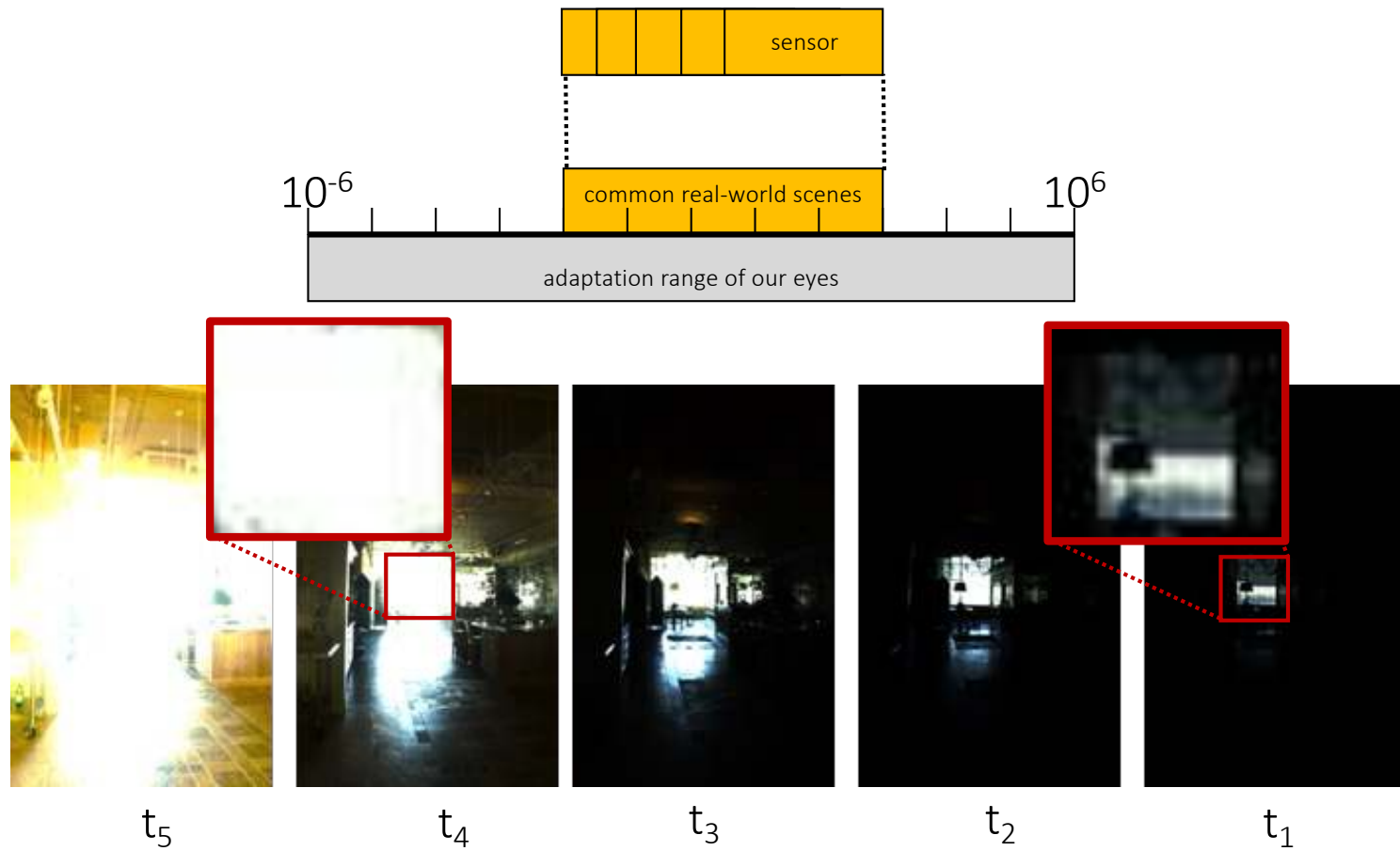


t_1

Multiple exposure bracketing: high noise



Multiple exposure bracketing: clipping



Linear (RAW) image formation model

Real scene radiance for image pixel (x,y) : $L(x, y)$

What is an expression for the image $I_{\text{linear}}(x,y)$ as a function of $L(x,y)$?

$$I_{\text{linear}}(x,y) = \text{clip}[t_i \cdot L(x,y) + \text{noise}]$$



t_5



t_4



t_3



t_2



t_1

Merging RAW (linear) exposure stacks

For each pixel:

1. Find “valid” images
2. Weight valid pixel values appropriately
3. Form a new pixel value as the weighted average of valid pixel values

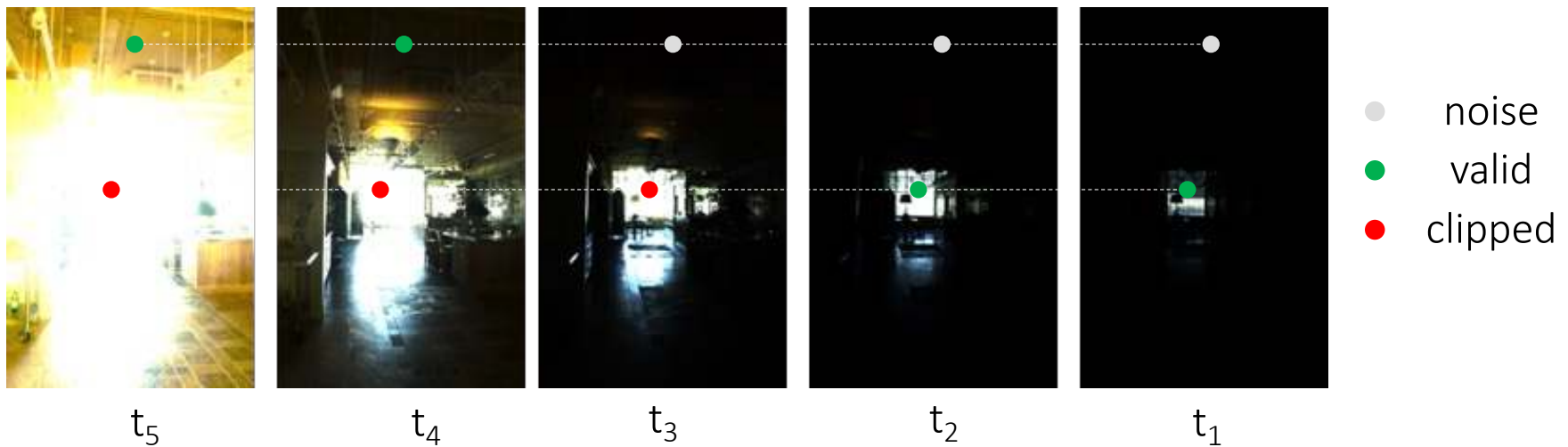
How would you
implement steps 1-2?



Merging RAW (linear) exposure stacks

For each pixel:

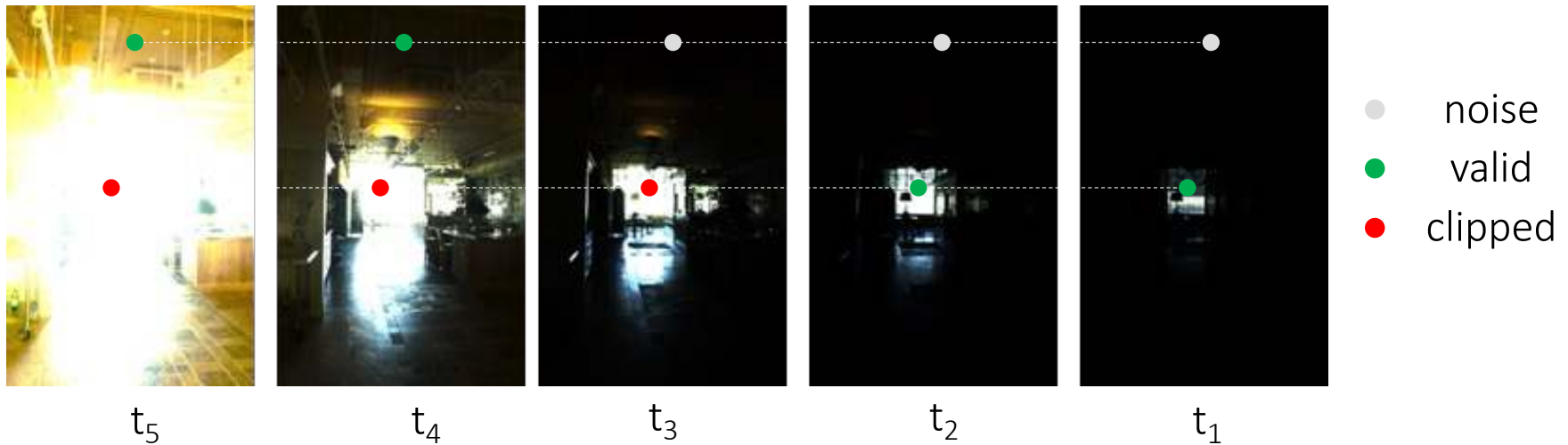
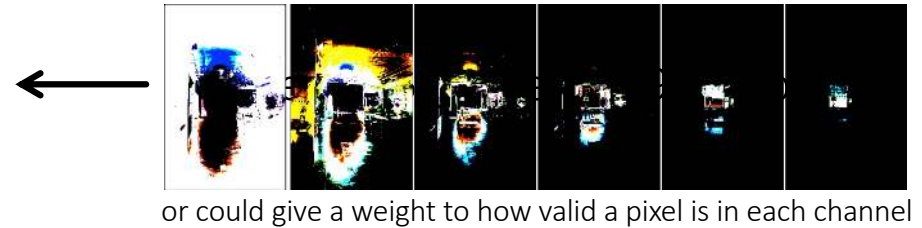
1. Find “valid” images ← (noise) $0.05 < \text{pixel} < 0.95$ (clipping)
2. Weight valid pixel values appropriately
3. Form a new pixel value as the weighted average of valid pixel values



Merging RAW (linear) exposure stacks

For each pixel:

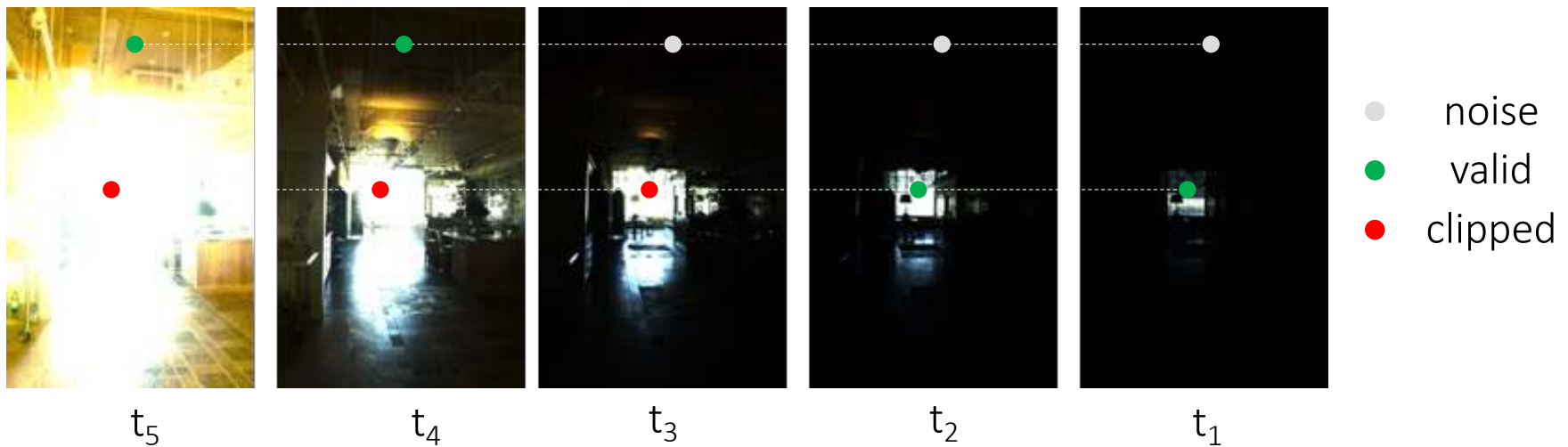
1. Find “valid” images
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3. Form a new pixel value as the weighted average of valid pixel values



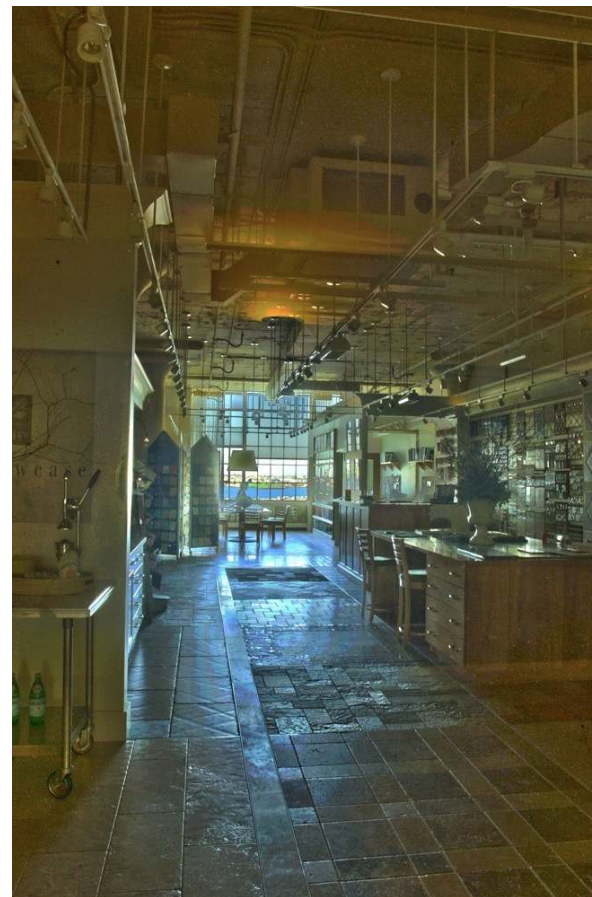
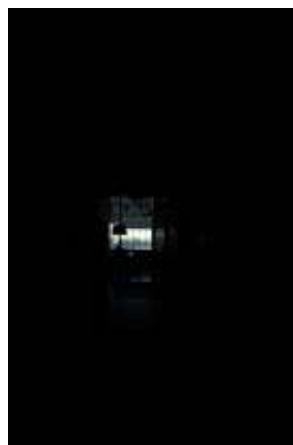
Merging RAW (linear) exposure stacks

For each pixel:

1. Find “valid” images $\longleftarrow$ (noise) $0.05 < \text{pixel} < 0.95$ (clipping)
2. Weight valid pixel values appropriately $\longleftarrow$ (pixel value) / t_i
3. Form a new pixel value as the weighted average of valid pixel values



Merging result (after tonemapping....coming next)



What if I cannot use raw?

Radiometric calibration

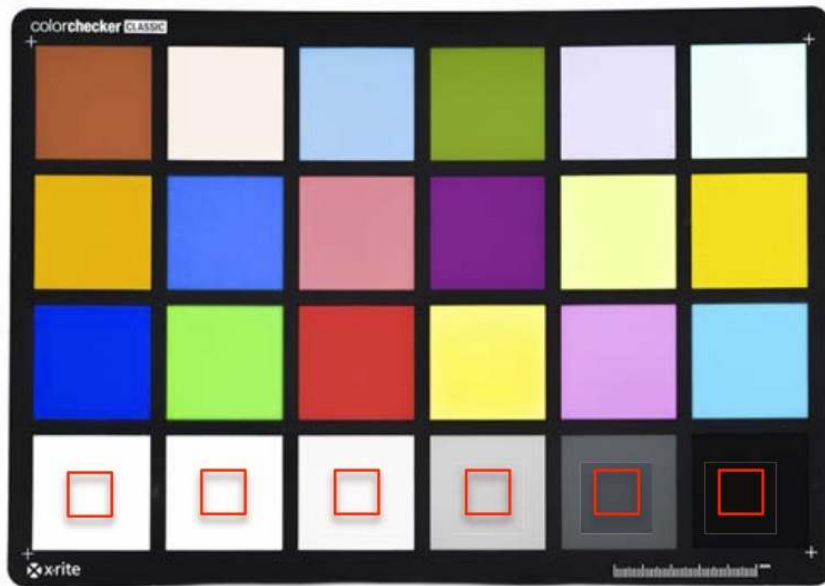
The process of measuring the camera's response curve. Can be done in two ways:

- Take images of scenes with different irradiance while keeping exposure the same.
- Take images under different exposures while keeping irradiance the same.

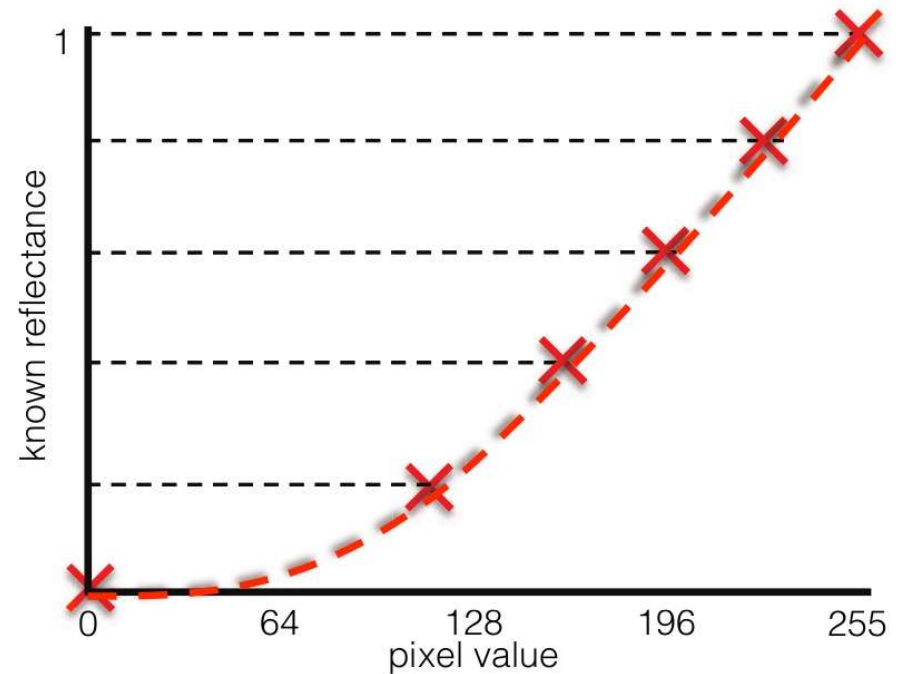
Same camera exposure, varying scene irradiance

Colorchecker: Great tool for radiometric and color calibration.

e.g. JPEG



Patches at bottom row have reflectance that increases linearly.



Different values correspond to patches of increasing reflected irradiance.

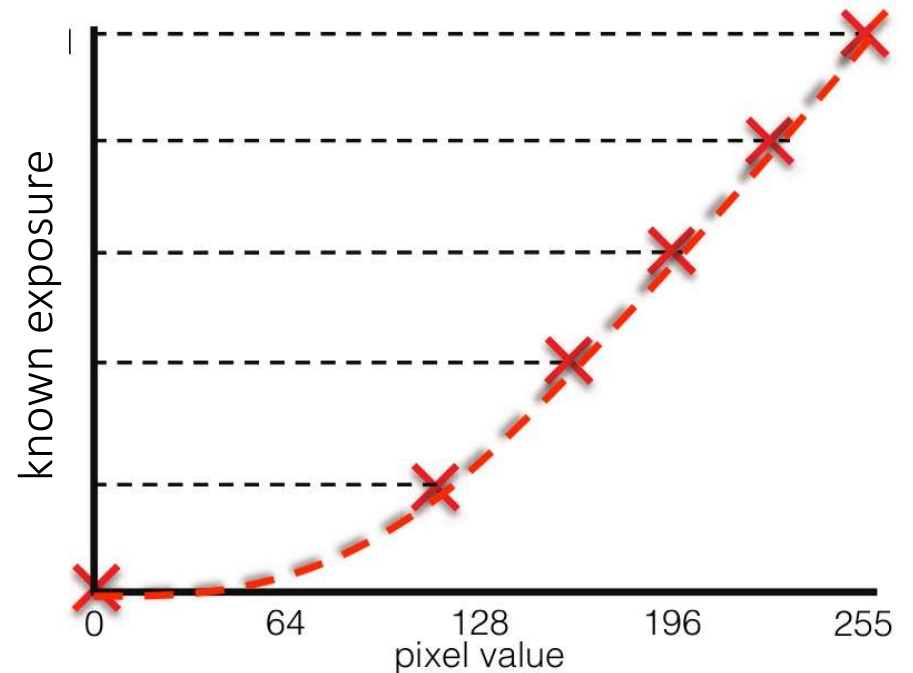
Same scene irradiance, varying camera exposure

Colorchecker: Great tool for white balancing and radiometric calibration.



All points on (the white part of) the target have the same reflectance.

e.g. JPEG

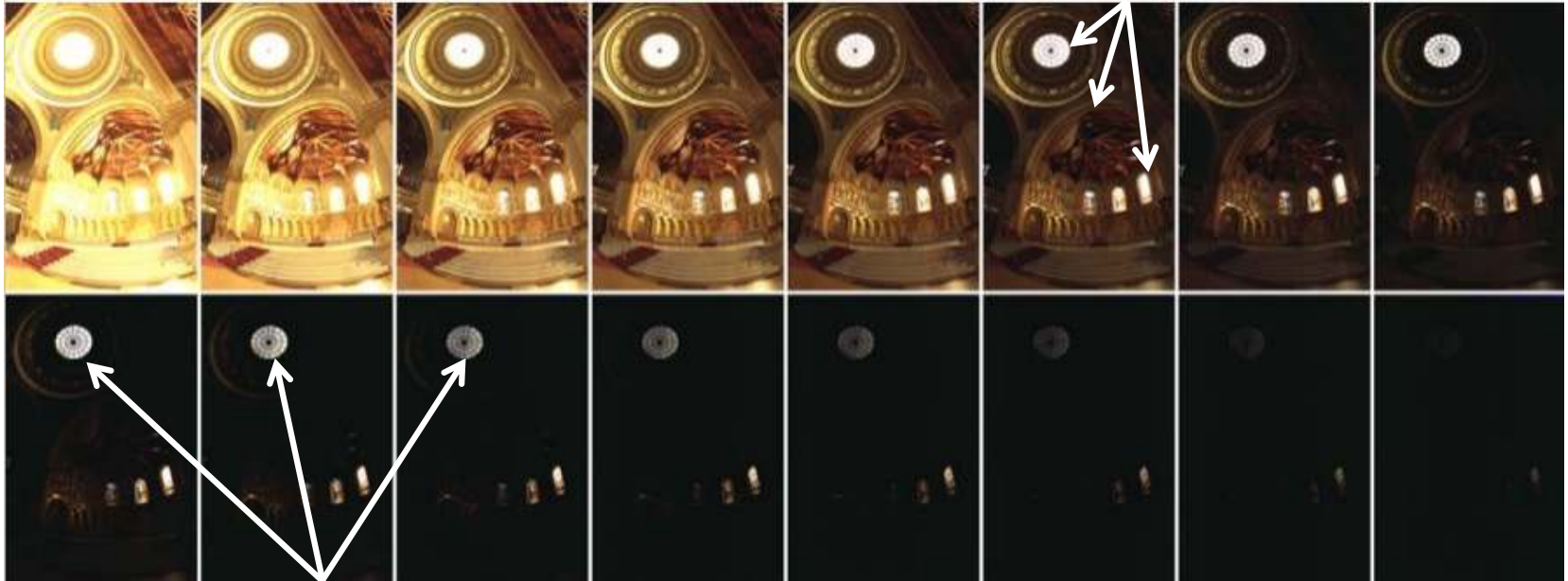


Different values correspond to images taken under increasing camera exposure.

Varying both scene irradiance and camera exposure

You can do this using the LDR exposure stack itself

Same scene irradiance, different camera exposure

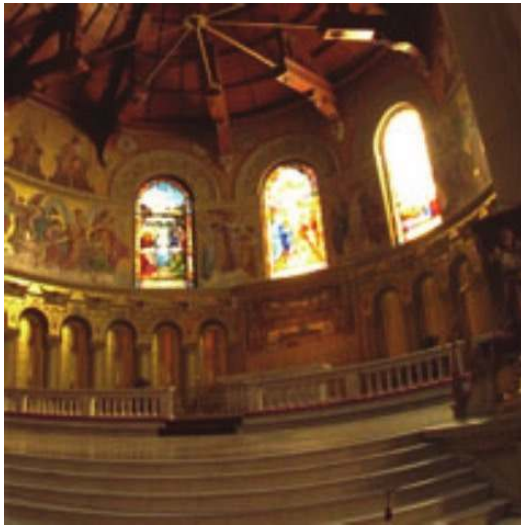


Same scene irradiance, different camera exposure

Non-linear image formation model

Real scene radiance for image pixel (x,y) : $L(x, y)$

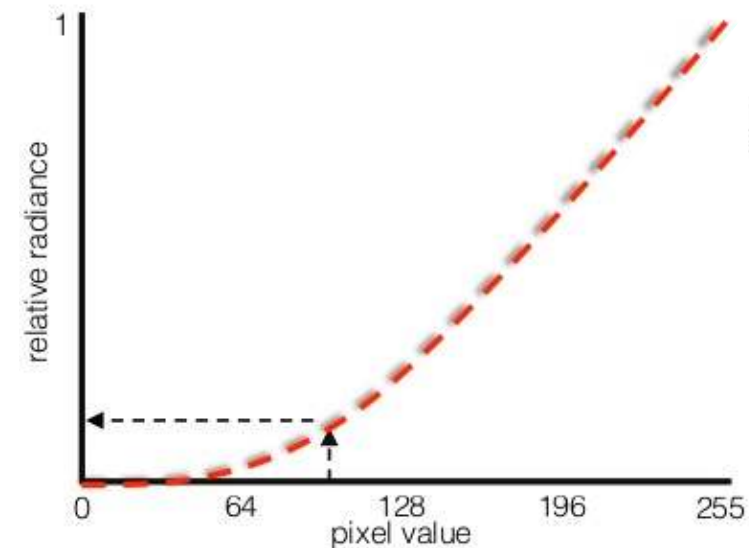
Exposure time: t_i



$$I_{\text{linear}}(x,y) = \text{clip}[t_i \cdot L(x,y) + \text{noise}]$$

$$I_{\text{non-linear}}(x,y) = f[I_{\text{linear}}(x,y)]$$

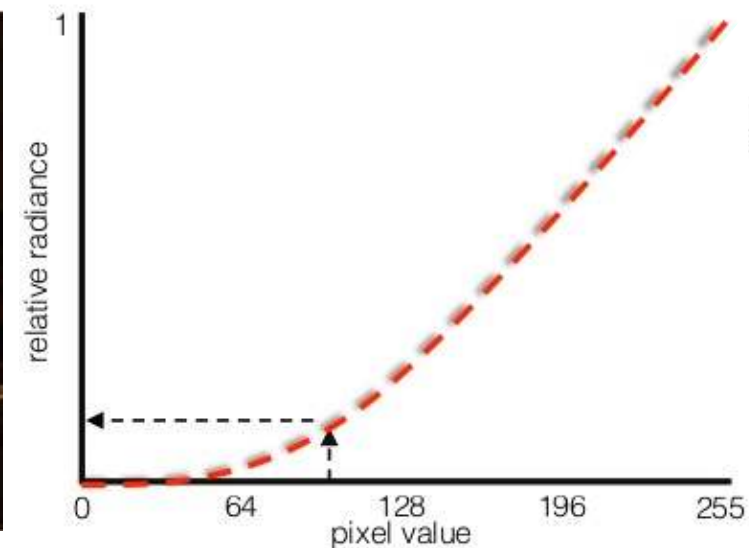
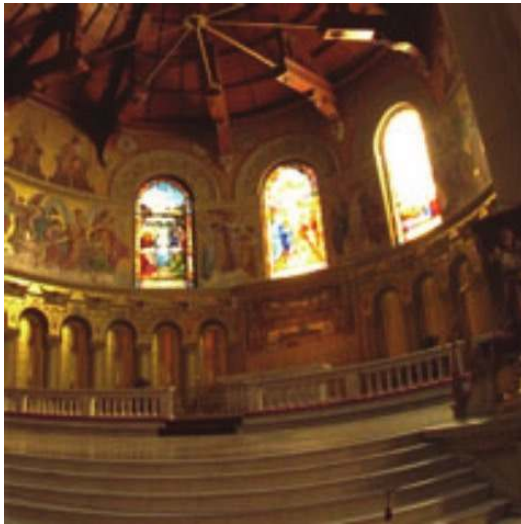
How would you merge the non-linear images into an HDR one?



Non-linear image formation model

Real scene radiance for image pixel (x,y): $L(x, y)$

Exposure time: t_i



$$I_{\text{linear}}(x,y) = \text{clip}[t_i \cdot L(x,y) + \text{noise}]$$

$$I_{\text{non-linear}}(x,y) = f[I_{\text{linear}}(x,y)] \quad I_{\text{est}}(x,y) = f^{-1}[I_{\text{non-linear}}(x,y)]$$

Use inverse transform to estimate linear image, then proceed as before

Linearization

$$I_{\text{non-linear}}(x,y) = f[I_{\text{linear}}(x,y)]$$



$$I_{\text{est}}(x,y) = f^{-1}[I_{\text{non-linear}}(x,y)]$$

Merging non-linear exposure stacks

1. Calibrate response curve
2. Linearize images

For each pixel:

- 
3. Find “valid” images ← (noise) $0.05 < \text{pixel} < 0.95$ (clipping)
 4. Weight valid pixel values appropriately ← $(\text{pixel value}) / t_i$
 5. Form a new pixel value as the weighted average of valid pixel values
- Same steps as in the RAW case.

Basic HDR approach

1. Capture multiple LDR images at different exposures
2. Merge them into a single HDR image
3. Store as a floating point 32- or 64-bit image

What are the problems with this approach?

Basic HDR approach

1. Capture multiple LDR images at different exposures
2. Merge them into a single HDR image
3. Store as a floating point 32- or 64-bit image

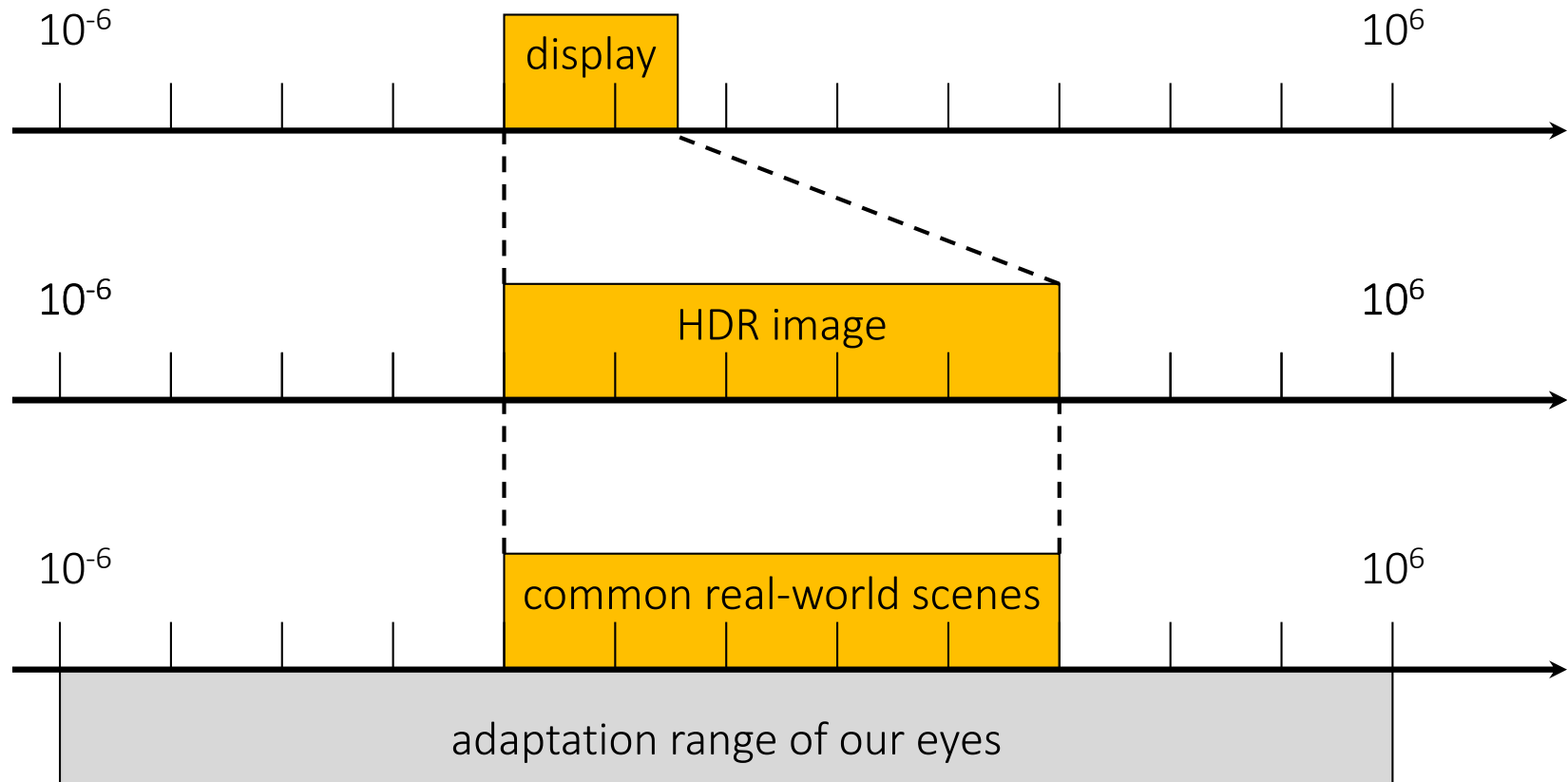
Problem: Very sensitive to movement

- Scene must be completely static
- Camera must not move

Most modern automatic HDR solutions include an alignment step before merging exposures

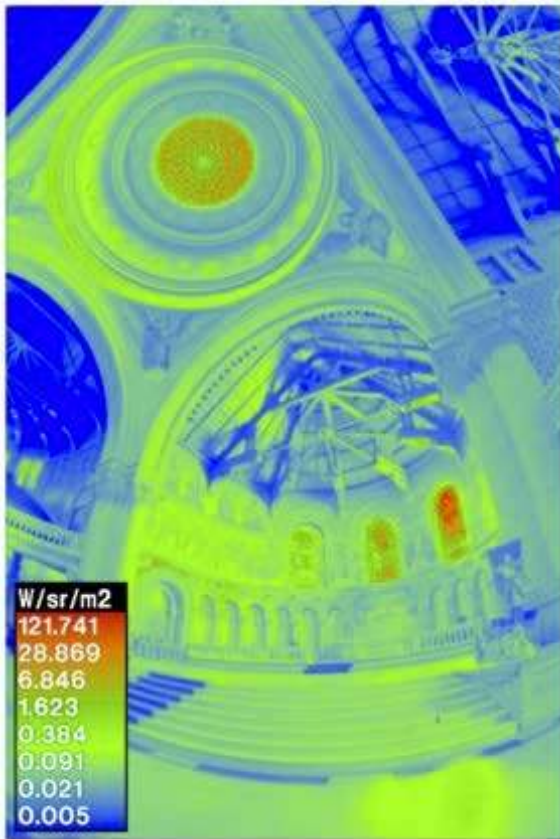
Tonemapping

How do we display our HDR images?



Linear scaling

Scale image so that maximum value equals 1

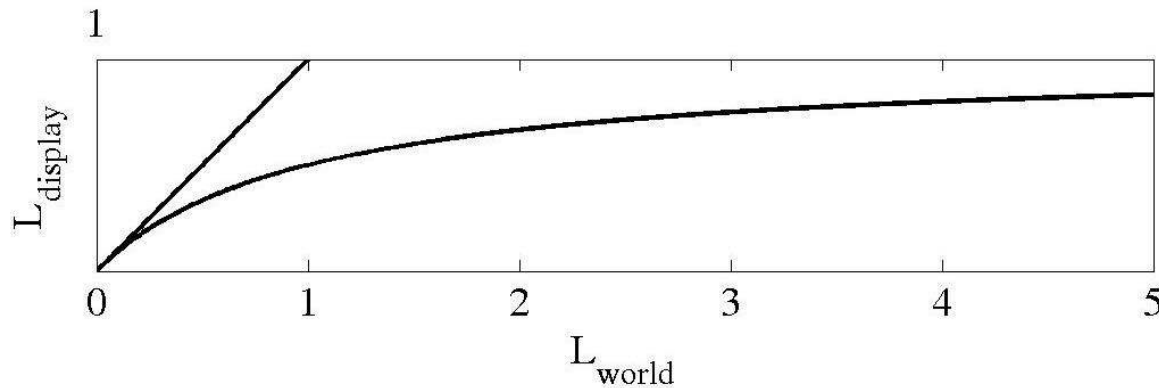


Can you think of something better?

Photographic tonemapping

Apply the same non-linear scaling to all pixels in the image so that:

- Bring everything within range $\rightarrow$ asymptote to 1
- Leave dark areas alone $\rightarrow$ slope = 1 near 0

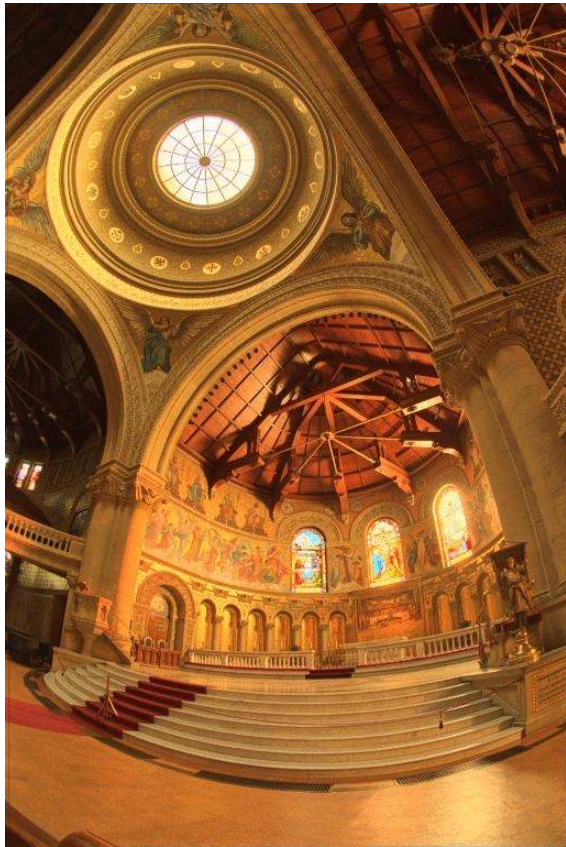


$$I_{display} = \frac{I_{HDR}}{1 + I_{HDR}}$$

(exact formula more complicated)

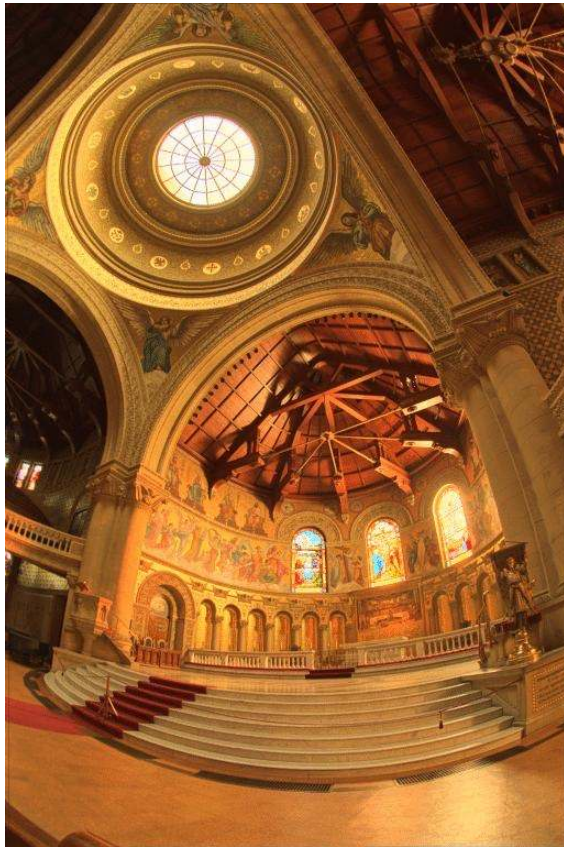
- Perceptually motivated, as it approximates our eye's response curve.

HDR Photographic tonemapping



Compare with LDR images

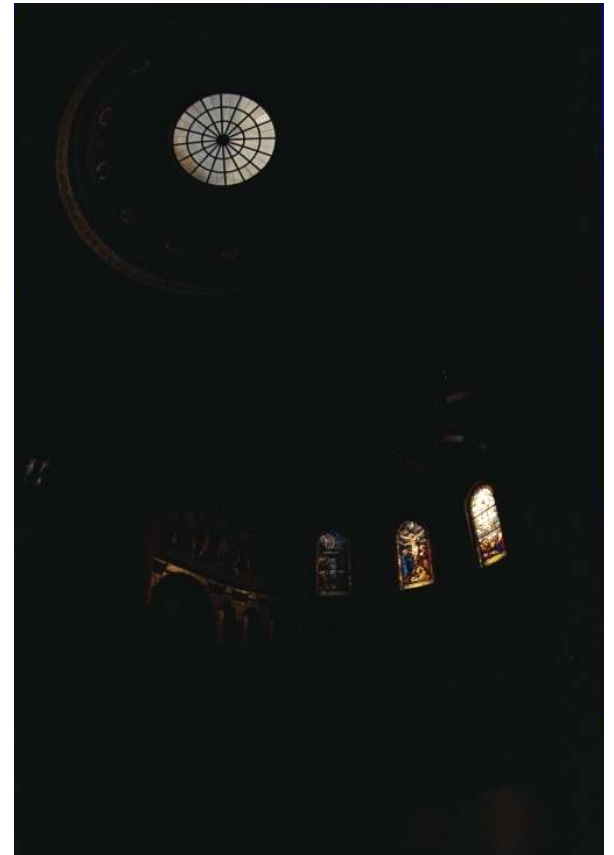
tonemapped



snapshot



linearly scaled HDR



Tonemapping

Scene has 100,000:1 dynamic range, we need to compress it down to 255:1



Tonemapping

If we tonemap all channels the same, colors are washed out



Can you think of a way to deal with this?

Intensity-only tonemapping

tonemap
intensity



leave color
the same



Comparison

Color now OK, but some details are washed out due to loss of contrast



Can you think of a way to deal with this?

Low-frequency intensity-only tonemapping

tonemap low-frequency
intensity component



leave high-frequency
intensity component
the same



leave color the same

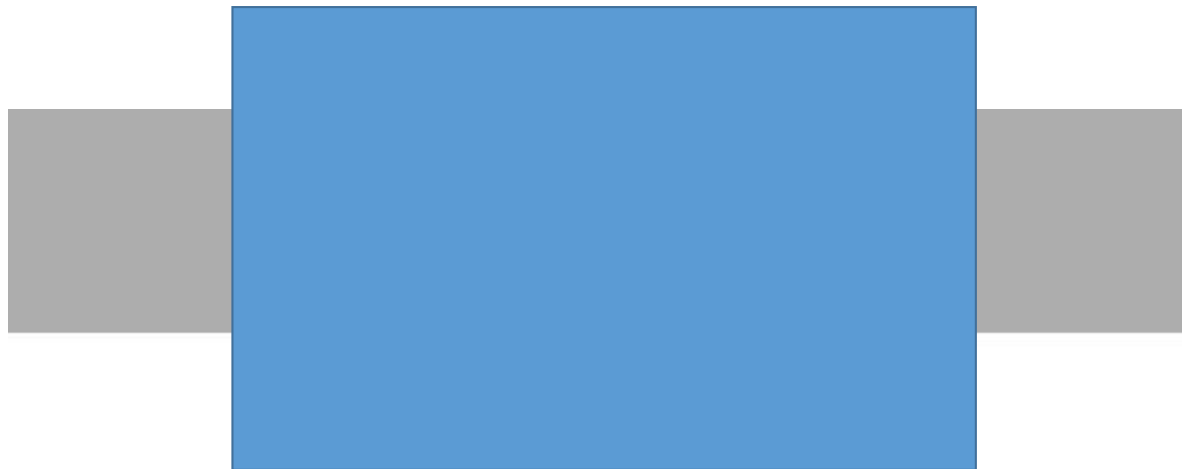


How would you implement this?

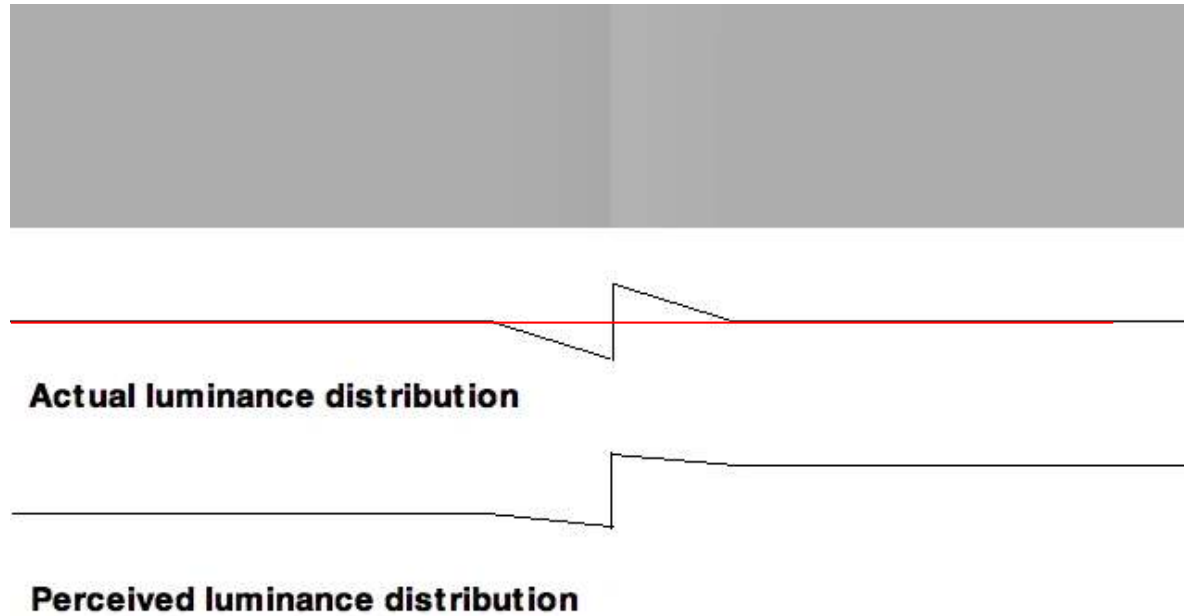
Cornsweet Illusion



Cornsweet Illusion



Cornsweet Illusion



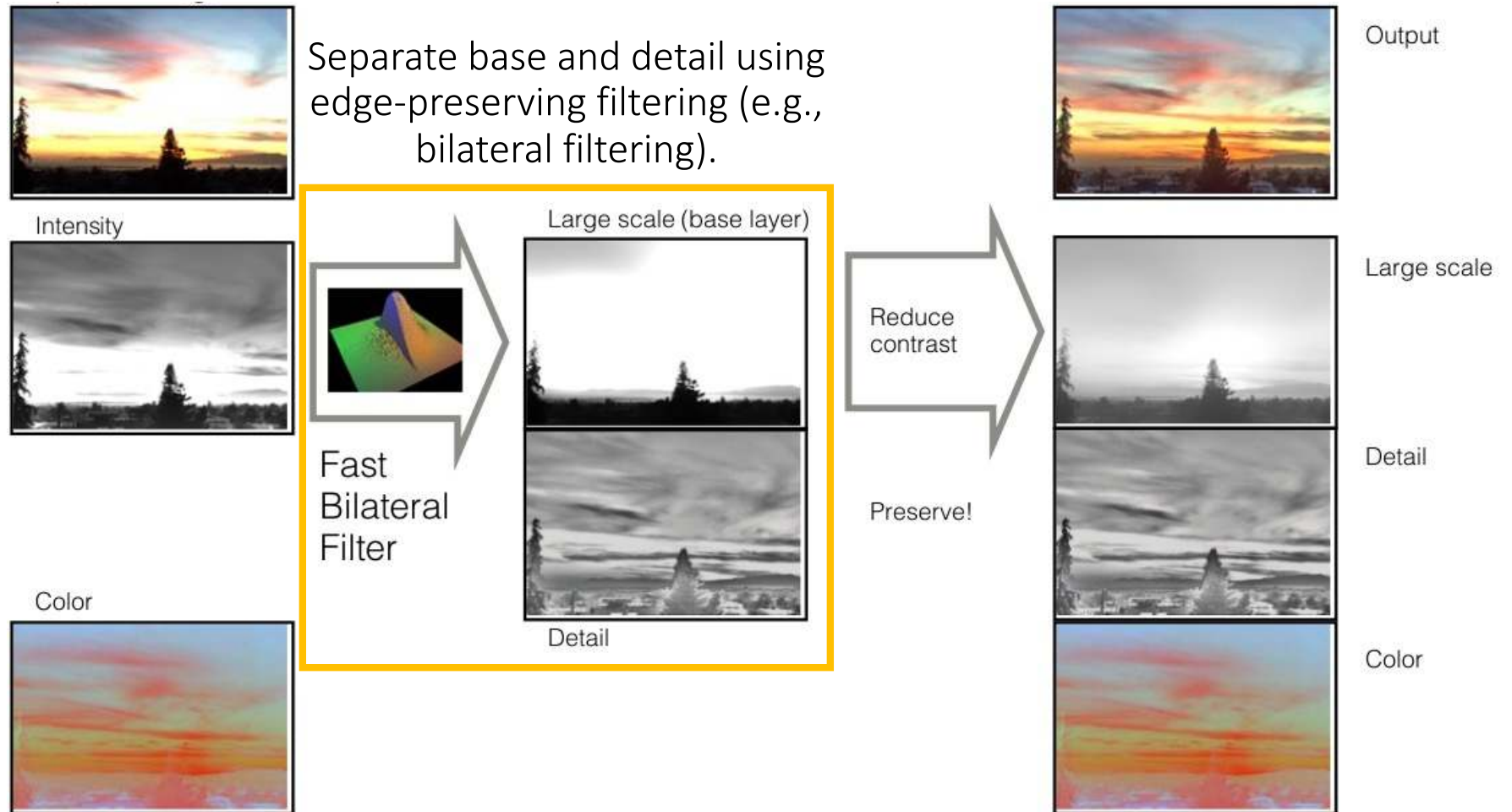
Comparison

We got nice color and contrast, but now we've run into the *halo* plague



Can you think of a way to deal with this?

Edge-aware filtering and tonemapping



Comparison

We fixed the halos without losing contrast



tone mapping on all channels separately



tone mapping on just the luminance



tone mapping Gaussian base layer of luminance



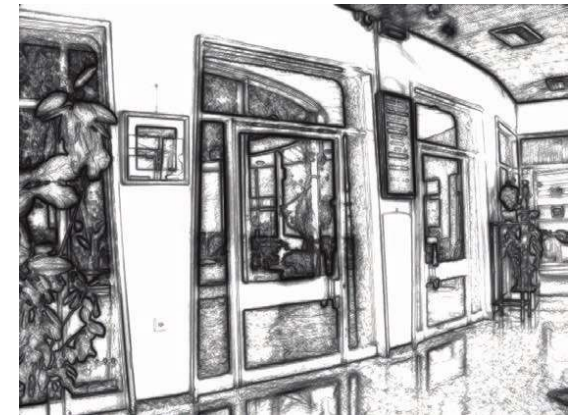
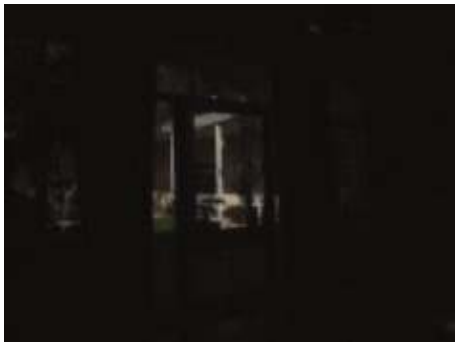
tone mapping bilateral base layer of luminance



Gradient-domain processing and tonemapping

Compute gradients, scale and merge them, then integrate (solve Poisson problem).

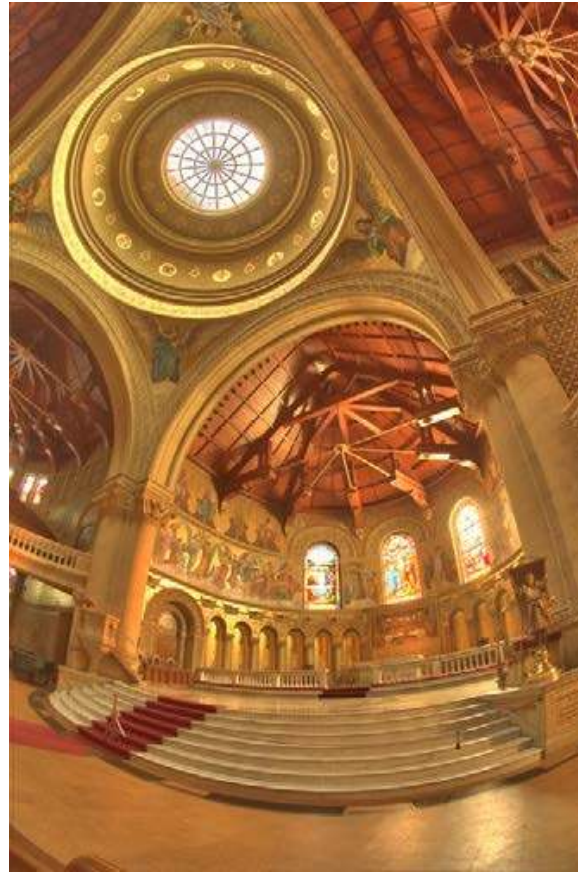
- More in later lecture.



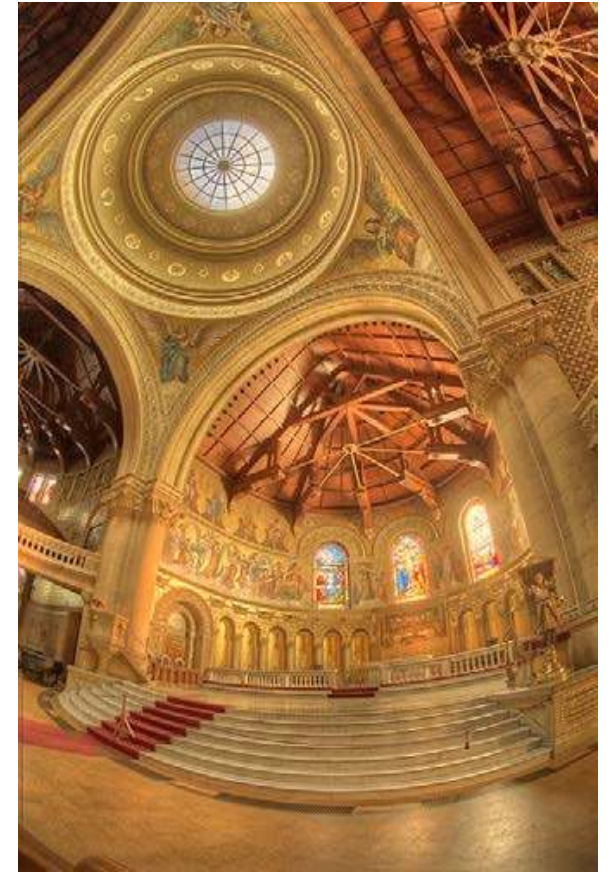
Comparison (which one do you like better?)



photographic



bilateral filtering



gradient-domain

Comparison (which one do you like better?)



photographic



bilateral filtering



gradient-domain

Comparison (which one do you like better?)



There is no ground-truth: which one looks better is entirely subjective



photographic

bilateral filtering

gradient-domain

A note of caution

- HDR photography can produce very visually compelling results.







A note of caution

- HDR photography can produce very visually compelling results.
- It is also a very routinely abused technique, resulting in awful results.









A note of caution

- HDR photography can produce very visually compelling results.
- It is also a very routinely abused technique, resulting in awful results.
- The problem typically is tonemapping, not HDR imaging itself.

A note about HDR today

- Most cameras (even phone cameras) have automatic HDR modes/apps.
- Popular-enough feature that phone manufacturers are actively competing about which one has the best HDR.
- The technology behind some of those apps (e.g., Google's HDR+) is published in SIGGRAPH and SIGGRAPH Asia conferences.

Burst photography for high dynamic range and low-light imaging on mobile cameras

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Jonathan T. Barron

Dillon Sharlet
Florian Kainz

Ryan Geiss
Jiawen Chen

Andrew Adams
Marc Levoy

Google Research



Figure 1: A comparison of a conventional camera pipeline (left, middle) and our burst photography pipeline (right) running on the same cell-phone camera. In this low-light setting (about 0.7 lux), the conventional camera pipeline underexposes (left). Brightening the image (middle) reveals heavy spatial denoising, which results in loss of detail and an unpleasantly blotchy appearance. Fusing a burst of images increases the signal-to-noise ratio, making aggressive spatial denoising unnecessary. We encourage the reader to zoom in. While our pipeline excels in low-light and high-dynamic-range scenes (for an example of the latter see figure 10), it is computationally efficient and reliably artifact-free, so it can be deployed on a mobile camera and used as a substitute for the conventional pipeline in almost all circumstances. For readability the figure has been made uniformly brighter than the original photographs.

Abstract

Cell phone cameras have small apertures, which limits the number of photons they can gather, leading to noisy images in low light. They also have small sensor pixels, which limits the number of electrons each pixel can store, leading to limited dynamic range. We describe a computational photography pipeline that captures, aligns, and merges a burst of frames to reduce noise and increase dynamic range. Our system has several key features that help make it robust and efficient. First, we do not use bracketed exposures. Instead, we capture frames of constant exposure, which makes alignment more robust, and we set this exposure low enough to avoid blowing out highlights. The resulting merged image has clean shadows and high bit depth, allowing us to apply standard HDR tone mapping methods. Second, we begin from Bayer raw frames rather than the demosaicked RGB (or YUV) frames produced by hardware Image Signal Processors (ISPs) common on mobile platforms. This gives us more bits per pixel and allows us to circumvent the ISP's unwanted tone mapping and spatial denoising. Third, we use a novel FFT-based alignment algorithm and a hybrid 2D/3D Wiener filter to denoise and merge the frames in a burst. Our implementation is built atop Android's Camera2 API, which provides per-frame camera control and access to raw imagery, and is written in the Halide domain-specific language (DSL). It runs in 4 seconds on device (for a 12 Mpix image), requires no user intervention, and ships on several mass-produced cell phones.

Keywords: computational photography, high dynamic range

Concepts: •Computing methodologies → Computational photography; Image processing;

1 Introduction

The main technical impediment to better photographs is lack of light. In indoor or night-time shots, the scene as a whole may provide insufficient light. The standard solution is either to apply analog or digital gain, which amplifies noise, or to lengthen exposure time, which causes motion blur due to camera shake or subject motion. Surprisingly, daytime shots with high dynamic range may also suffer from lack of light. In particular, if exposure time is reduced to avoid

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Questions?